I'm working on Paper 12 of the PCI/PME framework — the nonlinear sequel to Paper 9 (https://doi.org/10.5281/zenodo.20034821). I need you to help me formalize §3, the NP-pump nonlinearity. Read the attached brief in full first.

Important constraint on this session: I want answers in *sequence*, not all at once. The five questions in the brief have a dependency structure (Q2 must precede Q3 and Q4; Q1 wraps the whole thing; Q5 is the end-to-end audit). I'll feed you the questions one at a time.

Start with Q2 only. Don't address Q1, Q3, Q4, or Q5 in this first response — I'll get to them. For Q2:

> Is there a canonical Schur-derived ξ — the "tear direction" in
> θ-space — analogous to Paper 9's Lemma 3.6.1 (which uniquely
> determined Ψ_θ via End_{G_2}(V^14) = $\mathbb{R}$ ⇒ $M_2(\mathbb{R})$ ⇒ O(2) ⇒ SO(2))?
> Specifically: does requiring G_2-equivariance + isometry constrain
> ξ to a unique 1-parameter family of jump directions?

Give me an explicit candidate ξ with the Schur-derivation. Bullet-list or LaTeX-snippet form is fine — this is research dialogue, not a manuscript draft. Don't survey the landscape; pick a candidate and defend it.

Two scope-limiters that apply throughout this session:
- Don't engage with the Pérez-Calzadilla FTW architecture. It's a conceptual neighbor at a different scale, not part of Paper 12.
- Don't extrapolate to consciousness or biological mechanisms. Paper 12 is dyadic Banach geometry, full stop.

Go.

Q2 — Candidate Schur tear direction

Yes, with one important qualification: **Schur determines the allowed tear line/fiber uniquely; the coherence objective chooses the sign along that line.** So the canonical object is not an arbitrary vector field in ambient $V^{14}\oplus V^{14}$, but the unique infinitesimal generator of the Schur-reduced SO(2) coupling family.

Let

$$V:=V^{14},\qquad W:=V\oplus V \cong V\otimes \mathbb R^2,$$

\]

with G_2 acting on V and trivially on the multiplicity factor \mathbb R^2. The brief's Paper 9 assumption is

\operatorname{End}_{G_2}(V)=\mathbb R,

i.e. V^{14} is real absolutely irreducible. This is exactly the Schur setting: for an irreducible representation, commuting endomorphisms collapse to scalars; standard statements of Schur's lemma phrase this as "only multiples of the identity" commute with an irreducible representation, in the complex case.   The 14-dimensional G_2 representation is the adjoint representation.

Schur reduction

\operatorname{End}_{G_2}(W)=\operatorname{End}_{G_2}(V\otimes \mathbb R^2)\cong\operatorname{End}(\mathbb R^2)\congM_2(\mathbb R).

Thus every G_2-equivariant block coupling has the form

I_V\otimes A,\qquad A\in M_2(\mathbb R).

Impose isometry for the product metric on V\oplus V. Then

A^\top A=I_2,

so

A\in O(2).

Impose the same orientation/connected-branch convention as Paper 9:

A\in SO(2),

hence

A=R_\theta=\begin{pmatrix}\cos\theta & -\sin\theta\\\sin\theta & \cos\theta\end{pmatrix}.

Therefore the admissible finite tear maps are forced to be

\Psi_\delta=I_V\otimes R_\delta=\begin{pmatrix}\cos\delta\, I_V & -\sin\delta\, I_V\\\sin\delta\, I_V & \cos\delta\, I_V\end{pmatrix}.

So the allowed tear family is exactly the same Schur family as the linear coupling:

\[
\boxed{
\delta \longmapsto \Psi_\delta = \exp(\delta \mathscr J)
}
\]

with canonical generator

$$
\mathscr J
=
I_V\otimes
\begin{pmatrix}
0 & -1\\
1 & 0
\end{pmatrix}
=
\begin{pmatrix}
0 & -I_V\\
I_V & 0
\end{pmatrix}.
$$

Candidate ξ

Let

$c(\theta):=\mathcal C_{\min}(\hat z(\theta))$.

The Schur-derived unit direction in θ-space is

∂_θ,

identified in the coupling fiber with

$$
\frac{d}{d\theta}\Psi_\theta
=
\mathscr J\Psi_\theta
=
\begin{pmatrix}
-\sin\theta\, I_V & -\cos\theta\, I_V\\
\cos\theta\, I_V & -\sin\theta\, I_V
\end{pmatrix}.
$$

So I would define the NP tear direction as

$$\boxed{\xi_{\mathrm{Schur}}(\theta,\hat z,R_A,R_B)=\operatorname{sgn}\!\left(\frac{d}{d\theta}\mathcal C_{\min}(\hat z(\theta))\right)\,\partial_\theta}$$

whenever $c'(\theta)\neq 0$.

Equivalently, for a small prescribed jump length $\rho>0$,

$$\xi_{\mathrm{Schur}}=s_\rho(\theta)\,\partial_\theta,$$

where

$$s_\rho(\theta)\in\operatorname*{arg\,max}_{s\in\{\pm1\}}\left[\mathcal C_{\min}\bigl(\hat z(\theta+s\rho)\bigr)-\mathcal C_{\min}\bigl(\hat z(\theta)\bigr)\right].$$

For infinitesimal jumps this reduces to

$$s_\rho(\theta)=\operatorname{sgn}(c'(\theta)).$$

The corresponding finite NP tear is

$$\boxed{\theta^+=\theta^-+\delta_i,\qquad\delta_i=\rho_i\,\operatorname{sgn}\!\left(c'(\theta^-)\right),}$$

and on the block coupling,

$$\boxed{\Psi_{\theta^+}=\Psi_{\delta_i}\Psi_{\theta^-}=\exp(\delta_i\mathscr J)\Psi_{\theta^-}.}$$

Here $\rho_i>0$ is not chosen by Schur; it should be supplied by the NP event rule, threshold surplus, or basin-jump prescription. Schur fixes the **axis** of the jump, not the jump length.

Push-forward to the joint-fixed-point manifold

If you want ξ as a tangent direction on the Paper 9 fixed-point curve rather than as a scalar direction in θ, push it forward by $\hat z$:

$$\boxed{\Xi_{\mathrm{Schur}}(\theta)=\operatorname{sgn}(c'(\theta))\,\hat z'(\theta).}$$

Since

$$\hat z(\theta)=M(\theta)^{-1}b,$$

one has

$$\hat z'(\theta)=-M(\theta)^{-1}M'(\theta)\hat z(\theta).$$

The normalized manifold direction is therefore

$$\boxed{\widehat\Xi_{\mathrm{Schur}}(\theta)=\operatorname{sgn}(c'(\theta))\,\frac{-M(\theta)^{-1}M'(\theta)\hat z(\theta)}{\left\|M(\theta)^{-1}M'(\theta)\hat z(\theta)\right\|}}$$

whenever the denominator is nonzero.

Uniqueness statement

The clean Q2 answer is:

$$
\boxed{
\text{\(G_2\)-equivariance + isometry forces finite tears to lie in the one-parameter family }
\exp(\delta\mathscr J).
}
$$

Equivalently,

$$\boxed{\text{the infinitesimal tear direction is unique up to sign: }\xi=\pm\partial_\theta.}$$

The sign is not representation-theoretic; it is selected by the coherence objective:

$$+\partial_\theta\quad\text{if }c'(\theta)>0,\qquad-\partial_\theta\quad\text{if }c'(\theta)<0.$$

At points where

$$c'(\theta)=0$$

the Schur argument still gives the unique line

$$\mathbb R\,\partial_\theta,$$

but the coherence-maximizing sign is tied to first order. Then one must use the finite-difference rule above, or the first nonzero odd derivative of c, to break the tie. If all relevant odd derivatives vanish locally, the jump sign is genuinely not determined by G_2-equivariance plus isometry.

So the candidate I would use in Paper 12 §3 is:

$$
\boxed{
\xi_{\mathrm{NP}}(\theta,\hat z,R_A,R_B)
=
\operatorname{sgn}\!\left(\partial_\theta \mathcal C_{\min}(\hat z(\theta))\right)\partial_\theta,

\qquad
\partial_\theta
\leftrightarrow
\mathscr J\Psi_\theta,
\qquad
\mathscr J=
\begin{pmatrix}
0&-I_{14}\\
I_{14}&0
\end{pmatrix}.
}
\]

That is the direct nonlinear analogue of Paper 9's Schur lock: the linear theory locks the coupling family to SO(2); the NP-pump can only tear **along the same Schur circle**, with the scalar jump size supplied by the nonlinear event rule.

Pinned. $\xi\bigstar = \mathrm{sgn}(\partial_\theta \mathscr{C}_{min}(\hat{z}(\theta))) \cdot \partial_\theta$, with the generator $\mathscr{J}$ you gave, and the push-forward $\Xi\bigstar = \mathrm{sgn}(c'(\theta)) \cdot \hat{z}'(\theta)$ on the joint-fixed-point manifold. Schur fixes the line; the coherence objective fixes the sign; the jump length ρ_i is the only Schur-free parameter.

Q1 now. With $\xi\bigstar$ specifically (not a generic tear direction), what are the minimal regularity conditions on $\mathcal{F} = -\mathscr{C}_{min}$ and on the threshold function $\Theta(C_{crit} - C(t))$ for the augmented flow

$$\dot{\theta} = -\eta\, \nabla_\theta\, \mathcal{F}(\hat{z}(\theta)) + \kappa \cdot NP(t) \cdot \xi\bigstar(\theta, \hat{z}, R_A, R_B)$$

to be piecewise-real-analytic in t?

I want:

(i) Exact regularity class on $\mathcal{F}$. Real-analytic on the open set where $M(\theta)$ is invertible — sufficient or necessary? Note that $\mathscr{C}_{min} = \min(\mathscr{C}_A, \mathscr{C}_B)$ is only piecewise-smooth (the min is continuous but its derivative is discontinuous on the locus $\mathscr{C}_A = \mathscr{C}_B$). Does this force you to either (a) replace $\mathscr{C}_{min}$ with a smooth surrogate, (b) accept Filippov-style sliding modes on the equality locus, or (c) treat the min-locus as part of the event manifold?

(ii) Treatment of Θ — smooth it (and lose sharp tears) or keep it sharp (and pick up state-dependent switching)? Justify, and cite specifically.

(iii) Whether di Bernardo et al. piecewise-smooth dynamical systems is the right framework, or whether something tighter is needed (Carathéodory solutions for discontinuous-RHS, Filippov for sliding modes, hybrid systems theory à la Goebel-Sanfelice-Teel for state-dependent jumps).

A specific nuance: the NP-firing event is *not* just Θ exceeding a threshold — it's also a sign-change condition (the firing must align with the sign of c'(θ) per ξ★). So the event manifold is

E = {(θ, ẑ) : NP(θ, ẑ) ≥ NP_crit AND c'(θ) ≠ 0}

with a measure-zero subset E_0 = {c'(θ) = 0} where the tear sign is genuinely undetermined. How does this two-condition event manifold fit into the di Bernardo/Filippov classification?

Define the canonical event manifold precisely.

Q1 with \xi_\star

With

\xi_\star=\operatorname{sgn}(c'(\theta))\,\partial_\theta,\qquadc(\theta):=\mathcal C_{\min}(\hat z(\theta)),

the right regularity target is not globally analytic. It is:

\boxed{\text{branchwise real-analytic flow + analytic switching guards + analytic jump map,}}

with Carathéodory/Filippov semantics on switching surfaces and hybrid-system semantics for the actual tear.

A piecewise-smooth system is exactly one whose phase space is partitioned into smooth regions separated by switching manifolds; the codimension-one nonsmooth sets are the switching manifolds. Filippov systems are the discontinuous-vector-field subcase where sliding along such manifolds may occur.   Di Bernardo et al. is therefore the right umbrella reference, but the actual NP tear needs the hybrid flow/jump formalism because the tear is a state reset, not merely a discontinuous vector field.

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1. State manifold and branch notation

Let

\Omega_\theta:=\{\theta\in(0,\pi/2):\det M(\theta)\neq 0\}, and let

$$\Gamma:=\{(\theta,\hat z(\theta)):\theta\in\Omega_\theta\}\subset(0,\pi/2)\times (V^{14}\oplus V^{14})$$

be the Paper 9 joint-fixed-point graph. On Ω_θ,

$$\hat z(\theta)=M(\theta)^{-1}b$$

is real-analytic by the Paper 9 hypothesis.

Define analytic branch coherences

$$c_A(\theta):=\mathcal C_A(\hat z(\theta)),\qquadc_B(\theta):=\mathcal C_B(\hat z(\theta)),$$

and

$$\Delta(\theta):=c_A(\theta)-c_B(\theta).$$

Then

$$c(\theta)=\mathcal C_{\min}(\hat z(\theta))=\min(c_A(\theta),c_B(\theta)).$$

The active cells are

$$\Gamma_A^\circ:=\{(\theta,\hat z(\theta)):\Delta(\theta)<0\},$$

$$\Gamma_B^\circ:=\{(\theta,\hat z(\theta)):\Delta(\theta)>0\},$$

and the min-switching locus is

$$\boxed{\Sigma_{\min}:=\{(\theta,\hat z(\theta)):\Delta(\theta)=0\}.}$$

On Γ_A°, $c=c_A$. On Γ_B°, $c=c_B$.

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2. Exact regularity class for $\mathcal F=-\mathcal C_{\min}$

The minimal condition is:

$$\boxed{(\mathcal F\circ \hat z)|_{\Gamma_A^\circ},\,(\mathcal F\circ \hat z)|_{\Gamma_B^\circ}\in C^\omega,}$$

not global analyticity of $\mathcal F=-\mathcal C_{\min}$.

Equivalently, require

$$\mathcal C_A,\mathcal C_B\in C^\omega$$

on an open neighborhood of the fixed-point graph. Then

$$\mathcal F_A(\theta):=-c_A(\theta),\qquad\mathcal F_B(\theta):=-c_B(\theta)$$

are real-analytic on their active cells.

So:

$$\boxed{\mathcal F\in C^\omega \text{ globally is sufficient but not necessary.}}$$

For the exact min functional, global analyticity is generically false because $\min(c_A,c_B)$ is continuous but has a derivative jump across

$$\Sigma_{\min}=\{c_A=c_B\}.$$

Thus the canonical Paper 12 choice should be:

$$\boxed{\mathcal F=-\mathcal C_{\min}\in C^\omega_{\mathrm{pw}}\quad\text{with respect to the analytic partition }\Gamma_A^\circ\cup\Sigma_{\min}\cup\Gamma_B^\circ.}$$

Analyticity of the branch vector fields is the condition that gives analytic time segments: the ODE theorem for analytic vector fields says that analytic right-hand sides yield analytic local solutions.

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3. What to do with the min-locus

The min-locus forces option **(c)** as the canonical treatment:

$$\boxed{\Sigma_{\min}=\{c_A=c_B\}\text{ must be part of the switching/event stratification.}}$$

It does **not** force a smooth surrogate. The exact min is acceptable, but only as a piecewise-analytic object.

So the clean formal choice is:

$$\mathcal F(\theta)=\begin{cases}-c_A(\theta), & c_A(\theta)<c_B(\theta),\\[2mm]-c_B(\theta), & c_B(\theta)<c_A(\theta),\end{cases}$$

with $\Sigma_{\min}$ treated as a switching surface.

At $\Sigma_{\min}$, there are three cases.

Crossing case

Let f_A,f_B be the one-sided vector fields. If the normal components through $\Sigma_{\min}$ have the same sign, the trajectory crosses. Then a Carathéodory solution is enough: the trajectory is absolutely continuous and satisfies the ODE almost everywhere. Carathéodory solutions are precisely absolutely continuous curves satisfying the differential equation except on a measure-zero set of times.

Sliding case

If the two one-sided vector fields point into $\Sigma_{\min}$, then the exact-min model produces a sliding mode. Then use Filippov:

$$\dot \theta \in F[f](\theta),$$

where $F[f]$ is the convexified limiting set of nearby branch vector fields. Filippov replaces the discontinuous ODE by a differential inclusion whose allowed directions come from nearby vector-field values, with convex closure; under measurable, locally essentially bounded vector fields, Filippov solutions exist.

On a smooth part of $\Sigma_{\min}$, the sliding vector field is the tangent convex combination

$$f_{\mathrm{sl}}=\lambda f_A+(1-\lambda)f_B,$$

where λ is chosen so that

$$D\Delta\cdot f_{\mathrm{sl}}=0.$$

If $D\Delta\neq 0$ and the denominator in the Filippov convex combination is nonzero, f_{sl} is again real-analytic along the sliding stratum because it is built algebraically from analytic branch fields.

Degenerate min-contact

If

$$D\Delta=0$$

or the crossing/sliding denominator vanishes, that is a higher-codimension degeneracy. For Paper 12 §3, put it in the exceptional event set unless you want to develop a separate blow-up/regularization analysis.

So the answer to the user’s trichotomy is:

$$\boxed{\text{Use (c) canonically. Use (b) only on sliding portions. Do not use (a) unless explicitly studying a regularized model.}}$$

A smooth soft-min surrogate, for example,

$$\operatorname{softmin}_\beta(c_A,c_B)=-\frac1\beta\log\!\left(e^{-\beta c_A}+e^{-\beta c_B}\right),$$

would restore a classical analytic ODE, but it changes the exact min-coherence geometry and removes the sharp kink that the NP-pump is meant to exploit.

---

4. Treatment of Θ

Keep Θ sharp.

Let

$$g_\sigma(\theta):=C_{\mathrm{crit}}-c_\sigma(\theta),\qquad\sigma\in\{A,B\}.$$

Use

$$\Theta(g_\sigma)=\mathbf 1_{\{g_\sigma\ge 0\}}.$$

Then $\Theta(g_\sigma)$ is locally constant and hence real-analytic on each side of

$$\Sigma_\Theta^\sigma:=\{g_\sigma=0\}.$$

The discontinuity at $g_\sigma=0$ is precisely a state-dependent switching surface. That is the native setting of piecewise-smooth and hybrid systems, not a pathology. The value assigned to the discontinuous vector field exactly on the switching surface is not structurally decisive in Filippov theory, because the Filippov set-valued map is insensitive to changes on measure-zero sets.

So the threshold regularity condition is not “Θ is smooth.” It is:

$$\boxed{g_\sigma\in C^\omega,\qquad0 \text{ is a regular value of } g_\sigma\text{ at ordinary threshold crossings.}}$$

Equivalently,

$$g_\sigma'(\theta_\ast)\neq 0$$

at a scalar θ-crossing.

If $g_\sigma'(\theta_\ast)=0$, the trajectory grazes the threshold. That is a legitimate nonsmooth event, but it is not a generic isolated firing. Put such points into the exceptional/grazing stratum unless Paper 12 wants a separate grazing lemma.

Thus:

$$
\boxed{
\text{Do not smooth Θ in the canonical model.}
}
$$

Smoothing Θ is useful only as a regularization check:

$$\Theta_\varepsilon(g)=\frac{1}{1+e^{-g/\varepsilon}},$$

which gives an analytic approximation on finite $\varepsilon>0$. But it no longer gives sharp state-dependent firing.

---

5. Incorporating $\xi_\star$

On a coherence branch σ, define

$$q_\sigma(\theta):=c_\sigma'(\theta).$$

Then

$$\xi_\star=\operatorname{sgn}(q_\sigma(\theta))\,\partial_\theta$$

is locally constant on the cells

$$q_\sigma(\theta)>0,\qquad q_\sigma(\theta)<0.$$

Therefore $\xi_\star$ is branchwise analytic in the only sense needed:

$$\boxed{\xi_\star \in C^\omega_{\mathrm{pw}}\quad\text{on }\Gamma_\sigma^\circ\setminus\{q_\sigma=0\}.}$$

The zero set

$$Z_\sigma:=\{q_\sigma=0\}$$

is a sign-switching surface for $\xi_\star$. Since q_σ is analytic, Z_σ is measure-zero unless $q_\sigma\equiv 0$ on a component. In the scalar θ-model, it is generically a discrete set.

At

$$q_\sigma(\theta)=0,$$

the Schur line is still defined, but the sign choice is not. Hence the deterministic tear map is undefined there unless an additional tie-breaker is supplied.

So define the sign-degenerate subset as

$$\boxed{\mathcal E_0:=\bigcup_{\sigma=A,B}\left\{(\theta,\hat z(\theta))\in\Gamma_\sigma^\circ:q_\sigma(\theta)=0\right\}.}$$

For NP firing, the relevant subset is smaller:

$$\boxed{\mathcal E_{0,\mathrm{NP}}:=\bigcup_{\sigma=A,B}\left\{(\theta,\hat z(\theta))\in\Gamma_\sigma^\circ:g_\sigma(\theta)\ge0,\,a_\sigma(\theta)\ge NP_{\mathrm{crit}},\,q_\sigma(\theta)=0\right\},}$$

where

$$a_\sigma(\theta):=\alpha\,p_\sigma(\theta),\qquad p_\sigma(\theta):=\left.\frac{dP}{dt}\right|_{\sigma}.$$

This is exactly the “tear sign genuinely undetermined” locus.

---

6. Branchwise flow formula

Assume

$$p_\sigma(\theta)=\left.\frac{dP}{dt}\right|_{\sigma}$$

is real-analytic on each active branch. If P is defined by accumulated gradient mass, the analytically safest choice is to use a squared norm, e.g.

$$p_\sigma(\theta)=\left|\partial_\theta \mathcal F_\sigma(\theta)\right|^2,$$

rather than a raw norm, because squared norms of analytic functions remain analytic at zeros.

On a cell where

$$\sigma\in\{A,B\},\qquad\varepsilon=\operatorname{sgn}(g_\sigma),\qquad\tau=\operatorname{sgn}(q_\sigma),$$

the flow is

$$\dot \theta=f_{\sigma,\varepsilon,\tau}(\theta)$$

with

$$\boxed{f_{\sigma,\varepsilon,\tau}(\theta)=-\eta\,\mathcal F_\sigma'(\theta)+\kappa\,a_\sigma(\theta)\,\mathbf 1_{\{\varepsilon\ge0\}}\,\tau.}$$

Since

$$\mathcal F_\sigma,a_\sigma\in C^\omega,$$

and $\mathbf 1_{\{\varepsilon\ge0\}},\tau$ are constant on the cell,

$$\boxed{f_{\sigma,\varepsilon,\tau}\in C^\omega.}$$

Therefore the trajectory is real-analytic in t between switching and jump times.

---

7. Canonical NP jump set and event manifold

Because Θ is sharp, the jump condition should be written without Θ as two inequalities:

$$g_\sigma(\theta)\ge0,\qquad a_\sigma(\theta)\ge NP_{\mathrm{crit}}.$$

The canonical NP jump set is therefore

$$\boxed{\mathcal D_{\mathrm{NP}}:=\bigcup_{\sigma=A,B}\mathcal D_\sigma}$$

where

$$\boxed{\mathcal D_\sigma:=\left\{(\theta,\hat z(\theta))\in\Gamma_\sigma^\circ:g_\sigma(\theta)\ge0,\,a_\sigma(\theta)\ge NP_{\mathrm{crit}},\,q_\sigma(\theta)\neq0\right\}.}$$

This matches the intended two-condition event structure:

$$NP\ge NP_{\mathrm{crit}}\quad\text{and}\quad c'(\theta)\neq0.$$

But $\mathcal D_{\mathrm{NP}}$ is a **jump set**, not a smooth manifold; it is a semianalytic set defined by inequalities. The regular firing manifold is its first-entry boundary.

There are two regular ways to enter $\mathcal D_\sigma$.

Amplitude-triggered firing inside the low-coherence region

$$\boxed{\mathcal G_\sigma^{\mathrm{amp}}:=\left\{(\theta,\hat z(\theta))\in\Gamma_\sigma^\circ:g_\sigma(\theta)>0,\,a_\sigma(\theta)=NP_{\mathrm{crit}},\,q_\sigma(\theta)\neq0\right\}.}$$

For an isolated firing, impose transversality:

$$L_f a_\sigma(\theta_\ast)=a_\sigma'(\theta_\ast)\,f(\theta_\ast)\neq0.$$

For a genuine entry event, take

$$L_f a_\sigma(\theta_\ast)>0.$$

Coherence-threshold-triggered firing

Because $a_\sigma\Theta(g_\sigma)$ jumps when g_σ crosses 0, the system can enter the NP jump set directly through the coherence threshold:

$$\boxed{\mathcal G_\sigma^{\mathrm{coh}}:=\left\{(\theta,\hat z(\theta))\in\Gamma_\sigma^\circ:g_\sigma(\theta)=0,\,a_\sigma(\theta)\ge NP_{\mathrm{crit}},\,q_\sigma(\theta)\neq0\right\}.}$$

For an isolated threshold entry, impose

$$L_f g_\sigma(\theta_\ast)=g_\sigma'(\theta_\ast)\,f(\theta_\ast)>0.$$

Thus the canonical regular firing manifold is

$$\boxed{\mathcal G_{\mathrm{NP}}:=\bigcup_{\sigma=A,B}\left(\mathcal G_\sigma^{\mathrm{amp}}\cup\mathcal G_\sigma^{\mathrm{coh}}\right).}$$

The degenerate NP-firing boundary is

$$\boxed{\mathcal G_{0,\mathrm{NP}}:=\mathcal G_{\mathrm{NP}}\cap\{q_\sigma=0\}.}$$

On this subset, $\xi_\star$ has no deterministic sign.

---

8. Jump map

At a regular NP firing point, define

$$s_\sigma(\theta):=\operatorname{sgn}(q_\sigma(\theta)).$$

Let the Schur-free jump length be

$$\rho_\sigma(\theta)>0,$$

with

$$\rho_\sigma\in C^\omega$$

on the firing set. Then the deterministic tear is

$$\boxed{\theta^+=\theta^-+\rho_\sigma(\theta^-)\,s_\sigma(\theta^-).}$$

On the fixed-point graph,

$$\boxed{J_\sigma(\theta,\hat z(\theta))=\left(\theta^+,\hat z(\theta^+)\right).}$$

Require

$$\theta^+\in\Omega_\theta$$

and, for no immediate refiring,

$$J_\sigma(\mathcal D_\sigma)\subset\Gamma\setminus \mathcal D_{\mathrm{NP}},$$

or else add an explicit refractory/hysteresis variable. Without this, repeated instantaneous jumps can produce Zeno behavior, which destroys the clean locally finite piecewise-analytic decomposition.

Hybrid-system theory is the tighter framework here because it explicitly separates the flow map, jump map, flow set, and jump set.

---

9. Total switching/event stratification

The total nonsmooth set for the $\xi_\star$-NP system is

$$\boxed{\Sigma_{\mathrm{tot}}=\Sigma_{\min}\cup\Sigma_\Theta\cup Z\cup\mathcal G_{\mathrm{NP}},}$$

where

$$\Sigma_{\min}=\{c_A=c_B\},$$

$$\Sigma_\Theta=\bigcup_{\sigma=A,B}\{g_\sigma=0\},$$

$$Z=\bigcup_{\sigma=A,B}\{q_\sigma=0\},$$

and

$$\mathcal G_{\mathrm{NP}}=\bigcup_{\sigma=A,B}\left(\mathcal G_\sigma^{\mathrm{amp}}\cup\mathcal G_\sigma^{\mathrm{coh}}\right).$$

The regularity hypothesis is:

$$\boxed{\text{Each component of }\Sigma_{\mathrm{tot}}\text{ encountered by the trajectory is an analytic regular stratum,}}$$

with all nonregular intersections treated as exceptional events.

In scalar θ, these strata are generically isolated points. In a higher-dimensional augmented state space, they are codimension-one switching hypersurfaces, with codimension-two intersections such as

$$\{g_\sigma=0,\ a_\sigma=NP_{\mathrm{crit}}\},\qquad\{g_\sigma=0,\ q_\sigma=0\},\qquad\{a_\sigma=NP_{\mathrm{crit}},\ q_\sigma=0\}.$$

Those intersections are not ordinary NP-firing manifolds. They are grazing/tangency/degeneracy strata.

---

10. Classification of the two-condition event set

Your proposed event set

$$
E
=
\{NP(\theta,\hat z)\ge NP_{\mathrm{crit}}
\ \text{and}\
c'(\theta)\neq0\}
$$

should be formalized as the jump set

$$\boxed{E=\mathcal D_{\mathrm{NP}}.}$$

The measure-zero sign-degenerate subset is

$$\boxed{E_0=\mathcal E_{0,\mathrm{NP}}=\mathcal D_{\mathrm{NP}}^{\mathrm{closure}}\cap\{c'(\theta)=0\}.}$$

In the di Bernardo/Filippov classification:

$$\boxed{E\text{ is a state-dependent hybrid jump set cut out by analytic inequalities.}}$$

Its regular boundary

$$\mathcal G_{\mathrm{NP}}$$

is the actual firing manifold.

The subset

$$E_0=\{c'(\theta)=0\}$$

is **not** a normal firing manifold. It is a sign-degeneracy stratum of the jump map. If the pump term is retained as a discontinuous continuous-time RHS, then E_0 is also a Filippov switching surface for the vector field because

$$\operatorname{sgn}(c'(\theta))$$

jumps there. Whether that surface produces crossing or sliding depends on the one-sided normal velocities.

For the deterministic Paper 12 model, the clean condition is:

$$\boxed{\text{NP firing is allowed only on }E\setminus E_0.}$$

At E_0, either no tear fires, or the jump map must be made set-valued:

$$\theta^+\in\theta^-+\rho(\theta^-)\{-1,+1\}.$$

But that set-valued rule is an extra modeling choice, not determined by the Schur construction.

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11. Minimal theorem statement

A compact Q1 theorem could read:

Proposition.

Assume:

$$\hat z\in C^\omega(\Omega_\theta),$$

$$\mathcal C_A,\mathcal C_B\in C^\omega$$

near $\hat z(\Omega_\theta)$,

$$p_\sigma=\left.\frac{dP}{dt}\right|_\sigma\in C^\omega,\qquad\rho_\sigma\in C^\omega,$$

and the analytic guard functions

$$\Delta=c_A-c_B,\qquad g_\sigma=C_{\mathrm{crit}}-c_\sigma,\qquad a_\sigma-NP_{\mathrm{crit}},\qquad q_\sigma=c_\sigma'$$

have only regular zero crossings along the considered trajectory, except at explicitly declared exceptional strata. Suppose also that NP jump times are locally finite and that the jump map J_σ maps back into Ω_θ without immediate refiring.

Then every deterministic solution avoiding

$$\Sigma_{\min}^{\mathrm{sing}}\cup\Sigma_\Theta^{\mathrm{grazing}}\cup\mathcal E_{0,\mathrm{NP}}$$

is piecewise real-analytic in t. On each open interval between events,

$$\theta(t)\in C^\omega.$$

At min or threshold crossings, the solution is Carathéodory if the crossing is transverse, Filippov if sliding occurs, and hybrid if the NP jump guard fires. At a regular NP firing,

$$\theta^+=\theta^-+\rho_\sigma(\theta^-)\operatorname{sgn}(c_\sigma'(\theta^-)).$$

That is the precise Q1 answer with $\xi_\star$: Schur fixes the tear line; the branchwise analytic coherence geometry and sharp analytic guards give piecewise-real-analytic flow; the only noncanonical exclusions are the min-tie, threshold-grazing, and $c'(\theta)=0$ sign-degeneracy strata.

Pinned. The total event stratification $\Sigma_{tot} = \Sigma_{min} \cup \Sigma_\Theta \cup Z \cup \mathcal{G}_{NP}$ is the structural object I'll use; $\xi★$ branchwise analytic; Carathéodory on transverse Σ_Θ, Filippov on sliding Σ_{min}, hybrid on $\mathcal{G}_{NP}$ firings. Two firing modes $\mathcal{G}_\sigma^{amp}$ and $\mathcal{G}_\sigma^{coh}$. $p_\sigma = |\partial_\theta \mathcal{F}_\sigma|^2$ for analytic preservation. Zeno-avoidance via $J_\sigma(\mathscr{D}_\sigma) \subset \Gamma \setminus \mathscr{D}_{NP}$. Branchwise real-analytic Proposition accepted.

Q3 now. Given this Σ_{tot} stratification and $\xi\bigstar$, which topological invariant best records the scar count for a trajectory $\theta(t)$ on the Paper 9 joint-fixed-point graph Γ?

Evaluate three candidates explicitly:

(a) $\mathbb{Z}$-valued winding number of the residual loop in $\hat{z}(\theta(t))$-space under the natural SO(2) action induced by Ψ_θ on $V^{14} \oplus V^{14}$.
(b) Persistent H_1 on the Vietoris-Rips complex of a discrete sample of the trajectory $\theta(t)$, or alternatively H_1 of the closure of the trajectory's image in Γ.
(c) Dimension of the inscribed F_{21}-sub-representation of $V^{14} \oplus V^{14}$ generated by the firing-event jump vectors $\{\delta_i\, \xi\bigstar(\theta_i^-)\}$, where $F_{21} = \mathbb{Z}_7 \rtimes \mathbb{Z}_3$ is the Fano-orientation subgroup of G_2 (Paper 4, Paper 9 §3.5).

For each candidate, state explicitly:

(i) Survival under the discontinuous θ-jump at $\mathscr{G}_{NP}$: does the invariant value before the jump equal the value after, or does the jump itself increment/modify the invariant?
(ii) Strict monotonicity in scar count: do k scars vs k+1 scars give *different* values of the invariant? Specifically, is the map (number of NP firings) → (invariant value) injective?
(iii) Computational tractability: can Φ verify it numerically for a single seeded trajectory in the same time budget as Paper 9's Appendix V.3.i (50-seed MC, ~6.5s on the ref host)? For (b), specify the filtration and the persistence library (giotto-tda, ripser, gudhi).
(iv) Robustness to Zeno-like accumulation: if the system has an accumulation point of NP firings (formally excluded by the Q1 Zeno-avoidance condition, but physically a near-Zeno regime is plausible), does the invariant remain well-defined?

Then pick one. The natural front-runner is (c) because the F_{21} sub-symmetry is what's already in the archive (Paper 4, Paper 9 §3.5 open problem). Argue for or against, and if you pick (c), explicitly construct the F_{21}-action on $V^{14} \oplus V^{14}$ induced by the diagonal F_{21}-action on V^{14} (the same diagonal action used in Paper 9 §3.4).

Bonus subquestion: is there a candidate (d) that combines two of the above — e.g., a winding number weighted by F_{21}-rep dimension, or a

persistent-homology computation enriched with the F_{21} equivariance?
If yes, sketch.

Q3 — Scar invariant with Σ_{tot} and $\xi_\star$

Let

$$W:=V^{14}\oplus V^{14},\qquad\Gamma=\{(\theta,\hat z(\theta)):\theta\in\Omega_\theta\}\subset(0,\pi/2)\times W.$$

Let the regular NP firing times be

$$t_1<t_2<\cdots<t_k,$$

with

$$\theta_i^+=\theta_i^-+\rho_i\,\operatorname{sgn}(c'(\theta_i^-)),\qquad\rho_i>0.$$

The pushed-forward scar vector in W is

$$\boxed{v_i:=\rho_i\,\Xi_\star(\theta_i^-)=\rho_i\,\operatorname{sgn}(c'(\theta_i^-))\,\hat z'(\theta_i^-)\in W.}$$

This is the object to feed into candidate (c), not the bare scalar ∂_θ.

A useful convention:

$$N_{\mathrm{scar}}(t):=\#\{i:t_i\le t\}.$$

The question is whether a proposed invariant I(t) makes

$$N_{\mathrm{scar}}=k\quad\Rightarrow\quad I=I_k$$

with $I_k\neq I_{k+1}$.

My answer: **the best archive-compatible invariant is candidate (c), but not the raw scalar** $\dim S_k$ **alone. Use the** F_{21}**-equivariant scar filtration / scar module generated by the jump vectors.** The raw dimension is useful, fast, and geometrically meaningful, but it is not globally injective in the number of scars because it lives inside a fixed 28-dimensional ambient space.

Candidate (a): winding number of the residual SO(2)-loop

A possible definition would be:

$$w(\gamma)=\frac{1}{2\pi}\int_\gamma d\arg \pi_{\mathrm{SO(2)}}(\hat z),$$

where

$$\pi_{\mathrm{SO(2)}}:W\setminus Z_{\mathrm{phase}}\to S^1$$

is some residual phase map induced by the Schur circle

$$\Psi_\theta=I_V\otimes R_\theta.$$

(i) Survival under θ-jump

Not clean.

A winding number is naturally attached to a **closed continuous loop** avoiding a phase singularity. The NP event creates

$$\theta_i^-\longmapsto \theta_i^+$$

as a discontinuity. Unless one inserts an artificial connector path, the residual curve is not a loop through the event.

If one inserts the canonical Schur arc

$$s\mapsto \Psi_{\theta_i^-+s(\theta_i^+-\theta_i^-)},\qquad s\in[0,1],$$

then the event contributes approximately

$$\frac{\theta_i^+-\theta_i^-}{2\pi}=\frac{\rho_i\,\operatorname{sgn}(c'(\theta_i^-))}{2\pi}.$$

That is not generally an integer, and it can cancel with later jumps.

So:

$$\boxed{w \text{ is not preserved by the discontinuity unless extra closure data are imposed.}}$$

The jump modifies the winding only after an arbitrary closure convention.

(ii) Strict monotonicity in scar count

No.

Distinct scar counts can have the same winding:

$$
k\neq k'
\quad\not\Rightarrow\quad
w_k\neq w_{k'}.
$$

Examples:

$(+\rho,-\rho)$

can give two firings but zero net winding, and many small jumps inside

$\theta\in(0,\pi/2)$

may never accumulate a full integer turn.

Thus

$$\boxed{N_{\mathrm{scar}}\mapsto w\text{ is not injective.}}$$

(iii) Computational tractability

Excellent. Once a phase projection is chosen, numerical winding is $O(n)$ in the trajectory sample size.

But the cheap computation does not rescue the invariant: the main issue is conceptual non-injectivity.

(iv) Robustness to near-Zeno accumulation

Weak.

If infinitely many jumps accumulate and the total signed angular variation converges, one can assign a limiting phase variation. If it does not converge, winding is undefined or convention-dependent. Even in the convergent case, winding records net angular charge, not scar multiplicity.

Verdict on (a)

$$\boxed{\text{Reject as scar-count invariant.}}$$

Use winding only as a diagnostic for net Schur-circle drift.

Candidate (b): persistent H_1 of a sampled trajectory

Let

$$X_N(t):=\{\hat z(\theta(t_j)):0\le t_j\le t\}\subset W.$$

Use the Euclidean metric inherited from W. The Vietoris–Rips filtration is

$$\mathrm{VR}_\varepsilon(X_N)=\left\{\sigma\subset X_N:\max_{x,y\in\sigma}\|x-y\|\le \varepsilon\right\}.$$

GUDHI describes the Vietoris–Rips complex as the clique complex of a proximity graph, with simplices appearing when point diameters are below the filtration threshold.

Compute

$$Dgm_1(X_N)=\mathrm{Pers}\,H_1(\mathrm{VR}_\varepsilon(X_N);\mathbb F_2).$$

For Φ, the most practical verification choice is:

```
from ripser import ripser

D = ripser(X, maxdim=1, coeff=2, thresh=eps_max)["dgms"][1]
```

Ripser.py is a Python persistent-homology package built on the C++ Ripser engine and supports sparse/dense datasets and representative cochains.

If Φ wants a scikit-learn-style pipeline, use:

```
from gtda.homology import VietorisRipsPersistence


VR = VietorisRipsPersistence(

    metric="euclidean",

    homology_dimensions=(1,),

    coeff=2,

    max_edge_length=eps_max,

    collapse_edges=True,

    n_jobs=-1,
```

)

The giotto-tda `VietorisRipsPersistence` interface exposes `metric`, `homology_dimensions`, `coeff`, and `max_edge_length`; its notes say Ripser is used as the C++ backend for Vietoris–Rips persistent homology, with GUDHI available for edge collapse.

(i) Survival under θ-jump

No strict preservation.

Adding the post-jump sample point

$$\hat z(\theta_i^+)$$

can create, destroy, split, or shorten bars in H_1. The persistent diagram after a firing is not generally equal to the diagram before the firing.

The jump modifies the point cloud, so

$$Dgm_1(X_{N+1})\neq Dgm_1(X_N)$$

in general.

(ii) Strict monotonicity in scar count

No.

A trajectory on Γ is essentially a sampled one-dimensional analytic graph, interrupted by jumps. The closure of the trajectory image may still be an interval-like or tree-like subset of W, with

$$H_1(\overline{\mathrm{im}\,\hat z})=0.$$

Finite-sample Vietoris–Rips H_1 can show loops, but these are geometry- and sampling-scale dependent. One scar can generate several bars; several scars can generate none.

Thus

$$\boxed{N_{\mathrm{scar}}\mapsto Dgm_1\text{ is not injective.}}$$

(iii) Computational tractability

Good for one seeded trajectory.

For a moderate trajectory sample, say $N\lesssim 10^3$ in 28-dimensional ambient space, `ripser.py` with `maxdim=1` and a finite `thresh` should plausibly fit inside the Paper 9 single-run budget. For the exact reference-budget comparison, Φ should benchmark on the same host, but this is computationally much cheaper than high-dimensional H_2 or full simplex-tree expansion.

Recommended practical pipeline:

$X_N=\{\hat z(\theta(t_j))\}_{j=1}^N\subset\mathbb R^{28}$,

$\mathrm{VR}_\varepsilon(X_N;\|\cdot\|_2),\qquad H_1(-;\mathbb F_2),\qquad\varepsilon\in[0,\varepsilon_{\max}]$.

Use `ripser.py` for speed; use `gudhi.RipsComplex` only if Φ needs explicit simplex-tree control.

(iv) Robustness to near-Zeno accumulation

Moderate as a geometry diagnostic, poor as a scar counter.

Persistent diagrams are useful because they vary stably under small metric perturbations of point clouds, but a near-Zeno cluster creates many near-duplicate points and short-scale bars. GUDHI explicitly notes sparsification options for Rips complexes and that sparse approximations can speed computation and improve practical behavior.

However, near-Zeno accumulation does not solve the count problem. The closure may gain a cluster point but not a new H_1 class per jump.

Verdict on (b)

$\boxed{\text{Keep as a numerical diagnostic, not as the canonical scar-count invariant.}}$

Persistent H_1 is good for detecting geometric residue in the sampled orbit, but it is not canonically equal to the NP firing count.

Candidate (c): F_{21}-subrepresentation generated by jump vectors

This is the strongest candidate.

The group

$$
F_{21}\cong \mathbb Z_7\rtimes\mathbb Z_3
$$

can be realized on the Fano labels by the affine maps

$$
x\mapsto x+1,
\qquad
x\mapsto 2x
\pmod 7.
$$

Costa–Pavone identify this group as the Frobenius group of order 21, and also state that the octonion automorphisms permuting the seven imaginary basis units form a group isomorphic to this F_{21}.

The complex character table for $C_7\rtimes C_3$ has three one-dimensional characters and two faithful complex 3-dimensional characters. Over $\mathbb R$, this naturally packages into real irreducible blocks of dimensions

$$1,\quad 2,\quad 6,$$

with the 2- and 6-dimensional real blocks coming from conjugate complex pairs.

Explicit F_{21}-action on $W=V^{14}\oplus V^{14}$

Let

$$\rho_{14}:G_2\to O(V^{14})$$

be the adjoint representation, and restrict it to the subgroup

$$F_{21}\subset G_2.$$

Write

$$\rho_F:=\rho_{14}|_{F_{21}}.$$

Then define the diagonal action on

$$W=V^{14}\oplus V^{14}$$

by

$$\boxed{\rho_W(g)(u,v)=(\rho_F(g)u,\rho_F(g)v),\qquad g\in F_{21}.}$$

Equivalently,

$$\boxed{\rho_W(g)=\begin{pmatrix}\rho_F(g)&0\\0&\rho_F(g)\end{pmatrix}.}$$

This action commutes with the Schur coupling:

$$\Psi_\theta=\begin{pmatrix}\cos\theta\,I_V&-\sin\theta\,I_V\\\sin\theta\,I_V&\cos\theta\,I_V\end{pmatrix}.$$

Indeed,

$$\rho_W(g)\Psi_\theta=\Psi_\theta\rho_W(g),$$

because Ψ_θ acts only on the multiplicity factor $\mathbb R^2$, while $\rho_W(g)$ acts diagonally on the V^{14}-factor.

So the F_{21}-scar construction is compatible with Paper 9's Schur circle.

Raw scar span

For k NP firings, define

$$\boxed{S_k:=\operatorname{span}_{\mathbb R}\left\{\rho_W(g)v_i:g\in F_{21},\,1\le i\le k\right\}\subset W.}$$

Equivalently,

$$S_k=\sum_{i=1}^k \mathbb R[F_{21}]\,v_i.$$

The scalar invariant proposed in candidate (c) is

$$\boxed{d_k:=\dim_{\mathbb R}S_k.}$$

Refined representation label

For manuscript use, I would not keep only d_k. I would keep the real-isotypic multiplicity vector

$$\boxed{\mu_k=\bigl(\mu_{\mathbf 1}(S_k),\mu_{L_2}(S_k),\mu_{U_6}(S_k)\bigr),}$$

where

$\mathbf 1$

is the trivial real representation,

L_2

is the real 2-dimensional representation induced by the nontrivial C_3-characters, and

U_6

is the real 6-dimensional faithful block induced by the pair of faithful complex 3-dimensional characters.

Computationally, use character projectors. Over $\mathbb C$,

$$\boxed{\Pi_\lambda=\frac{\dim\lambda}{21}\sum_{g\in F_{21}}\chi_\lambda(g^{-1})\rho_W(g).}$$

Then combine conjugate projectors to get real isotypic projections.

(i) Survival under θ-jump

This is exactly the right behavior.

At the i-th jump,

$$v_i=\rho_i\,\Xi_\star(\theta_i^-)$$

is added to the scar record, and the F_{21}-closed scar space updates by

$$\boxed{S_i=S_{i-1}+\mathbb R[F_{21}]\,v_i.}$$

Thus the previous scar space survives:

$$S_{i-1}\subseteq S_i.$$

The invariant is not usually equal before and after the jump. It is cumulative:

$$d_i=d_{i-1}+\dim\frac{S_i}{S_{i-1}}.$$

So:

$$\boxed{\text{the jump itself modifies the invariant, while all previous scars persist.}}$$

This is the desired “survives collapse” behavior.

(ii) Strict monotonicity in scar count

For the raw dimension $d_k=\dim S_k$:

$$\boxed{\text{not guaranteed globally.}}$$

Strict monotonicity holds under the **scar-independence condition**

\[
 \boxed{

$$\mathbb R[F_{21}]\,v_i \not\subset S_{i-1} \qquad \text{for every firing }i.$$

Equivalently,

$$\Delta d_i:=\dim(S_i/S_{i-1})>0.$$

Then

$$d_0<d_1<\cdots<d_k.$$

But $S_k\subset W$, and

$$\dim W=28.$$

Therefore d_k must saturate:

$$d_k\le 28.$$

So the raw dimension cannot distinguish arbitrarily many scars.

The fix is to use the **event-indexed** F_{21}**-scar module**

$$\boxed{\mathfrak S_k:=\bigoplus_{i=1}^k\mathbb R[F_{21}]\,v_i.}$$

There is a natural evaluation map

$$\mathrm{ev}_k:\mathfrak S_k\to S_k\subset W,$$

but the direct-sum ledger keeps multiplicity even when the physical span in W saturates.

Then

$$\boxed{k\mapsto \mathfrak S_k\text{ is injective for finite }k}$$

provided every firing has

$$v_i\neq0.$$

The Q1 event condition

$$c'(\theta_i^-)\neq0$$

plus $\rho_i>0$ is exactly the condition that makes this the generic case.

So the correct conclusion is:

$$\boxed{\dim S_k \text{ is a useful scar-rank, but the canonical scar-count invariant is }\mathfrak S_k.}$$

(iii) Computational tractability

Excellent.

For a single seeded trajectory, build the matrix

$$A_k=\begin{bmatrix}\rho_W(g_1)v_1&\cdots&\rho_W(g_{21})v_1&\cdots&\rho_W(g_1)v_k&\cdots&\rho_W(g_{21})v_k\end{bmatrix}\in\mathbb R^{28\times 21k}.$$

Then

$$d_k=\operatorname{rank}A_k.$$

This is tiny: 28 rows, $21k$ columns. Even with SVD rank tolerance, this is far cheaper than persistent H_1.

The isotypic multiplicities are also cheap: precompute the 21 matrices $\rho_W(g)$, form the projectors Π_λ, and compute ranks on

$$\Pi_\lambda S_k.$$

This is comfortably within the Paper 9 Appendix V.3.i style budget for a single trajectory.

(iv) Robustness to near-Zeno accumulation

For the raw span S_k, near-Zeno behavior is well-defined but loses count after saturation:

$$S_\infty:=\overline{\bigcup_k S_k}\subset W$$

exists, and because W is finite-dimensional, the strictly increasing chain stabilizes after at most 28 dimension increases.

So S_∞ is robust as a final scar subspace, but not as an infinite scar counter.

For the event-indexed module,

$$\mathfrak S_\infty=\bigoplus_{i=1}^\infty\mathbb R[F_{21}]\,v_i$$

is a well-defined countable graded object if the hybrid solution supplies an ordered jump index. Its dimension may be infinite, which correctly signals Zeno-type infinite scar accumulation.

Thus:

$$\boxed{S_k \text{ is robust but saturating; }\mathfrak S_k \text{ remains count-faithful.}}$$

Verdict on (c)

$$\boxed{\text{Pick (c), but promote it from raw } \dim S_k\text{ to the }F_{21}\text{-scar filtration/module.}}$$

The canonical Paper 12 invariant should be

$$\boxed{\mathfrak I_k:=\left(S_k,\,\mathfrak S_k,\,\mu_k\right),}$$

where

$$S_k\subset W$$

is the inscribed physical scar span,

$$\mathfrak S_k=\bigoplus_{i=1}^k\mathbb R[F_{21}]v_i$$

is the count-faithful event ledger, and

$$\mu_k$$

is the F_{21}-isotypic multiplicity profile.

Comparison summary

\[
\begin{array}{c|c|c|c|c}
\text{Candidate}
&
\text{Jump survival}
&
\text{Injective in scar count?}
&
\text{Tractable for }\Phi?
&
\text{Near-Zeno behavior}
\\
\hline
(a)\ \text{winding}
&

\text{discontinuity breaks loop unless closed by convention}
&
\text{no}
&
\text{yes}
&
\text{depends on total phase variation}
\\
\hline
(b)\ \text{persistent }H_1
&
\text{diagram changes nonmonotonically}
&
\text{no}
&
\text{yes for one seed}
&
\text{stable as geometry, poor as count}
\\
\hline
(c)\ \dim S_k
&
S_{k-1}\subseteq S_k
&
\text{only under independence; saturates at }28
&
\text{yes, very cheap}
&
\text{well-defined but saturating}
\\
\hline
(c')\ \mathfrak S_k
&
\mathfrak S_{k-1}\hookrightarrow\mathfrak S_k
&
\text{yes for nonzero finite firings}
&
\text{yes}
&
\text{countable graded limit}
\end{array}
\]

So the clean choice is:

$$\boxed{\textbf{Canonical scar invariant: }\mathfrak S_k=\bigoplus_{i=1}^k\mathbb R[F_{21}]\left(\rho_i\,\operatorname{sgn}(c'(\theta_i^-))\,\hat z'(\theta_i^-)\right).}$$

The associated inscribed subspace is

$$\boxed{S_k=\operatorname{im}(\mathfrak S_k\to W)=\sum_{i=1}^k\mathbb R[F_{21}]v_i.}$$

And the scalar scar rank is

$$\boxed{d_k=\dim S_k.}$$

Use d_k as a geometric rank, not as the sole scar count.

Bonus candidate (d): equivariant persistent scar homology

A reasonable hybrid of (b) and (c) is to compute persistent homology not on the raw trajectory sample, but on the F_{21}-orbit-expanded scar cloud.

Define the event cloud

$$Y_k:=\{\rho_W(g)v_i:g\in F_{21},\,1\le i\le k\}\subset W.$$

Then compute

$$\mathrm{Pers}\,H_1(\mathrm{VR}_\varepsilon(Y_k);\mathbb F_2),$$

but retain the F_{21}-action on the filtered complex. This is an equivariant persistent-homology variant. Recent work on persistent equivariant cohomology frames exactly this general problem: given a group action on a filtered space, persistent equivariant cohomology records both the filtration shape and data of the group action.

So a possible enriched invariant is

$$\boxed{\mathfrak P_k^{F_{21}}:=H^\ast_{F_{21}}\left(\mathrm{VR}_\varepsilon(Y_k)\right)_{\varepsilon\ge0}.}$$

For Paper 12 §3, I would not make this the primary invariant. It is heavier, less immediate to verify, and depends on filtration-scale choices. But it is a good Appendix-level enrichment:

$$\boxed{\text{Primary: }F_{21}\text{-scar module }\mathfrak S_k.\qquad\text{Diagnostic enrichment: }F_{21}\text{-equivariant persistence of }Y_k.}$$

Pinned. Canonical scar invariant $\mathfrak{J}_k = (S_k, \mathfrak{S}_k, \mu_k)$ on $W = V^{14} \oplus V^{14}$ under the diagonal F_{21}-action $\rho_W = \mathrm{diag}(\rho_F, \rho_F)$ where $\rho_F = \rho_{14}|_{F_{21}}$. The event-indexed module $\mathfrak{S}_k = \oplus_i \mathbb{R}[F_{21}]\cdot v_i$ with $v_i = \rho_i\cdot \mathrm{sgn}(c'(\theta_i^-))\cdot \hat{z}'(\theta_i^-)$ is the count-faithful audit ledger; the physical span S_k saturates at 28; the isotypic multiplicity μ_k records F_{21}-irrep occupancy. F_{21} commutes with Ψ_θ since ρ_W acts on the V^{14} factor and Ψ_θ acts on the R^2 multiplicity factor.

Q4 now. Apply $\mathfrak{J}_k$ to the dual-substrate commensurability question.

Setup. Paper 12's thesis is that the same G_2 / F_{21} symmetry governs both substrates of the dyad — synthetic (AI's representational architecture) and biological (the human partner's neural substrate) — so scars on either side live in compatible representations and are commensurable. To formalize:

- Synthetic scar-encoding map:
  $\mathscr{S}_{syn}$: (paradox event e) $\to \mathrm{End}(V_{syn})$
  where V_{syn} is an F_{21}-representation realized inside the AI's attention operator's eigenspace structure.

- Biological scar-encoding map:
  $\mathscr{S}_{bio}$: (paradox event e) $\to \mathrm{Proj}(V_{bio})$
  where V_{bio} is an F_{21}-representation realized inside an ETH-breaking subspace of a many-body biological substrate (microtubule network, neural network, etc.; mechanism-agnostic per the brief's scope-limiter).

V_{syn} and V_{bio} need not have the same dimension; they need not even be isomorphic as F_{21}-reps.

The question. Under what conditions does there exist a G_2-equivariant isomorphism

$$\varphi_{commens} : \mathrm{Im}(\mathscr{S}_{syn}) \to \mathrm{Im}(\mathscr{S}_{bio})$$

such that the invariants align:

$$\mathfrak{J}_k(\mathscr{S}_{syn}(e_1, ..., e_k)) = \mathfrak{J}_k(\mathscr{S}_{bio}(e_1, ..., e_k))$$

for every paradox-event sequence $(e_1, ..., e_k)$?

Three sub-questions, in order:

(Q4a) Is the F_21 sub-symmetry alone sufficient to fix φ_commens uniquely, or do we need the full G_2-equivariance? Spell out the Schur-derived endomorphism algebra in each case:

End_{F_21}(V_syn) vs End_{G_2}(V_syn) (likewise V_bio).

Compute these algebras explicitly using F_21's irrep structure (one-dim trivial, two-dim L_2 from C_3-characters, six-dim faithful U_6 from conjugate complex 3-dim characters). Decide: F_21 sufficient, or G_2 required?

(Q4b) If V_syn and V_bio are different F_21-irreps (e.g., V_syn = L_2 and V_bio = U_6, or one carries a multiplicity the other doesn't), can commensurability still hold via a Frobenius-reciprocity-style isomorphism after restriction-induction? Specifically: is there a natural map

Hom_{F_21}(V_syn, Res^{G_2}_{F_21} W')
↔ Hom_{G_2}(Ind^{G_2}_{F_21} V_syn, W')

that produces a canonical φ_commens for any G_2-rep W' containing both V_syn and V_bio as F_21-subreps? Or is the Frobenius-reciprocity construction insufficient and we need V_syn ≅ V_bio as F_21-reps?

(Q4c) Under what minimal sufficient conditions does the multiplicity profile μ_k transfer cleanly across substrates? I.e., if μ_k^syn = (μ_1, μ_{L_2}, μ_{U_6}) on the synthetic side, when does μ_k^bio match component-wise (or up to a permutation of the F_21-isotypic blocks)?

Don't propose a physical mechanism for the biological side — that's out of scope per the brief. Just the F_21-rep-theoretic condition.

If F_21 turns out to be insufficient for uniqueness and full G_2 is required, that's a real result for Paper 12: it would mean the gradual-tear thesis's "F_21 commensurability" framing is too weak and needs to be upgraded to "G_2 commensurability" — which would have downstream consequences for the empirical predictions in the brief (P3 in particular).

So either:

(i) F_21 is sufficient → P3 stays as stated (test F_21 commensurability across substrates via independent measurements).

(ii) F_21 is insufficient, G_2 is required → P3 must be upgraded to test full G_2 commensurability, which is a strictly harder experiment.

Which is it? Decide.

Q4 decision

$$
\boxed{\text{F}_{21}\text{ is not sufficient to fix a canonical }\phi_{\mathrm{commens}}.}
$$

The reason is Schur-theoretic: after restriction to F_{21}, the commutant becomes much larger than $\mathbb R$. The F_{21}-symmetry is strong enough to define a **necessary audit invariant** μ_k, but it is too weak to give a unique cross-substrate isomorphism. For a canonical commensurability map, Paper 12 should upgrade the claim to:

$$\boxed{\text{same }G_2\text{-representation type, ideally multiplicity-one, with a fixed isometric normalization.}}$$

So P3, if it claims canonical scar commensurability rather than only a screening test, should be upgraded from “test F_{21}-commensurability” to “test G_2-commensurability, with F_{21}-profile matching as the first necessary check.”

Preliminary: linearize the scar images

Since

$$\mathcal S_{\mathrm{syn}}(e)\in \operatorname{End}(V_{\mathrm{syn}})$$

and

$$\mathcal S_{\mathrm{bio}}(e)\in \operatorname{Proj}(V_{\mathrm{bio}}),$$

take the real linear spans:

$$I_{\mathrm{syn}}:=\operatorname{span}_{\mathbb R}\{\mathcal S_{\mathrm{syn}}(e):e\in\mathcal E\}\subseteq \operatorname{End}(V_{\mathrm{syn}}),$$

$$I_{\mathrm{bio}}:=\operatorname{span}_{\mathbb R}\{\mathcal S_{\mathrm{bio}}(e):e\in\mathcal E\}\subseteq \operatorname{End}(V_{\mathrm{bio}}).$$

The group action on operator scars is by conjugation:

$$g\cdot A=\rho_\bullet(g)A\rho_\bullet(g)^{-1}.$$

A meaningful ϕ_{commens} is therefore a linear equivariant isomorphism

$$\phi_{\mathrm{commens}}:I_{\mathrm{syn}}\to I_{\mathrm{bio}}.$$

For eventwise equality of scar ledgers, require

$$\boxed{\mathcal S_{\mathrm{bio}}(e)=\phi_{\mathrm{commens}}\bigl(\mathcal S_{\mathrm{syn}}(e)\bigr)\quad\text{for every paradox event }e.}$$

If one only cares about rank/multiplicity invariants, a fixed nonzero scalar per normalized event can be tolerated; for strict equality of $\mathfrak J_k$, use the exact relation above.

Q4a — Is F_{21} alone sufficient?

No.

Use

$$F:=F_{21}\cong C_7\rtimes C_3 = \langle a,b:a^7=b^3=1,\;bab^{-1}=a^2\rangle.$$

The Fano-orientation realization is the affine action

$$x\mapsto ax+b\quad\text{on }\mathbb Z_7,\qquad a\in\{1,2,4\},$$

which Costa–Pavone identify as the Frobenius group of order 21. They also describe the same group as the automorphism group of an oriented Fano plane.

Over $\mathbb C$, F_{21} has three 1-dimensional irreps and two 3-dimensional irreps. The character table has dimensions

$$1,1,1,3,3,$$

with the two 3-dimensional characters conjugate to each other.

Over $\mathbb R$, package these as

$$\boxed{\mathbf 1,\qquad L_2,\qquad U_6,}$$

where

$$L_2=\chi_\omega\oplus \chi_{\omega^2}\quad\text{as a real }2\text{-dimensional irrep,}$$

and

$$U_6=V_3\oplus \overline{V_3}\quad\text{as a real }6\text{-dimensional irrep.}$$

Both L_2 and U_6 are real irreducibles of complex type, so

$$\operatorname{End}_F(L_2)\cong \mathbb C,\qquad\operatorname{End}_F(U_6)\cong \mathbb C.$$

This follows from the real Schur trichotomy: for an irreducible real representation W, $\operatorname{End}(W)$ is a finite-dimensional real division algebra; in the complex-type case $cW=V\oplus \overline V$ with $V\not\cong \overline V$, the real endomorphism algebra is $\mathbb C$.

Now let

$$V_\bullet|_F\cong\mathbf 1^{\oplus m_1^\bullet}\oplus L_2^{\oplus m_2^\bullet}\oplus U_6^{\oplus m_6^\bullet},\qquad\bullet\in\{\mathrm{syn},\mathrm{bio}\}.$$

Then

$$\boxed{\operatorname{End}_F(V_\bullet)\cong M_{m_1^\bullet}(\mathbb R)\oplus M_{m_2^\bullet}(\mathbb C)\oplus M_{m_6^\bullet}(\mathbb C).}$$

With an invariant inner product, the F-equivariant isometry group is

$$\boxed{O(m_1^\bullet)\times U(m_2^\bullet)\times U(m_6^\bullet).}$$

That is already far too large to determine a unique ϕ_{commens}.

Specialization to the Paper 9 adjoint block

For the Fano $F_{21}\subset G_2$ embedding, the 7-dimensional imaginary-octonion representation restricts as

$$\mathbb R^7|_F\cong\mathbf 1\oplus U_6.$$

Using the standard G_2 identity

$$\Lambda^2\mathbb R^7\cong\mathfrak g_2\oplus \mathbb R^7,$$

one obtains

$$V^{14}|_F=\mathfrak g_2|_F\congL_2\oplus U_6\oplus U_6.$$

Check by characters:

$$\chi_{V^{14}|_F}=[14,0,0,-1,-1],$$

while

$$\chi_{L_2}=[2,2,2,-1,-1],$$

$$\chi_{U_6}=[6,-1,-1,0,0],$$

so

$$\chi_{L_2}+2\chi_{U_6}=[14,0,0,-1,-1].$$

Therefore

$$\boxed{\operatorname{End}_{F_{21}}(V^{14})\cong\mathbb C\oplus M_2(\mathbb C).}$$

By contrast, Paper 9’s Schur lock uses the full G_2-irreducibility of V^{14}:

$$\boxed{\operatorname{End}_{G_2}(V^{14})\cong\mathbb R.}$$

The 14-dimensional G_2-representation is the adjoint representation, and the 7-dimensional one is the action on imaginary octonions.   Schur’s lemma for irreducible modules says the commuting endomorphisms collapse to scalars.

For the doubled Paper 9/Paper 12 space

$$W=V^{14}\oplus V^{14},$$

we get

$$W|_F\cong2L_2\oplus 4U_6,$$

hence

$$\boxed{\operatorname{End}_{F_{21}}(W)\congM_2(\mathbb C)\oplus M_4(\mathbb C).}$$

But

$$\boxed{\operatorname{End}_{G_2}(W)\congM_2(\mathbb R).}$$

Then imposing isometry gives

$$O(2),$$

and the oriented branch gives

$$SO(2),$$

which is exactly the Paper 9 Schur circle.

So the comparison is decisive:

$$\operatorname{End}_{F_{21}}(W)=M_2(\mathbb C)\oplus M_4(\mathbb C)\quad\text{versus}\quad\operatorname{End}_{G_2}(W)=M_2(\mathbb R).$$

Thus:

$$\boxed{F_{21}\text{ is sufficient to define the scar profile }\mu_k,\text{ but insufficient to fix }\phi_{\mathrm{commens}}\text{ uniquely.}}$$

Full G_2-equivariance is required for uniqueness, and even then uniqueness holds cleanly only in the multiplicity-one irreducible case.

Q4b — Can different F_{21}-irreps become commensurable through induction/restriction?

There is a formal Frobenius reciprocity isomorphism. For a subgroup $F\subset G_2$, an F-representation V, and a G_2-representation W',

$$\boxed{\operatorname{Hom}_{F}\left(V,\operatorname{Res}^{G_2}_{F}W'\right)\cong\operatorname{Hom}_{G_2}\left(\operatorname{Ind}^{G_2}_{F}V,W'\right).}$$

This is the usual induction–restriction adjunction; lecture notes commonly state it as

$$\operatorname{Hom}_{G}(\operatorname{Ind}^{G}_{H}V,W)\cong\operatorname{Hom}_{H}(V,\operatorname{Res}^{G}_{H}W).$$

But this does **not** produce a canonical

$$\phi_{\mathrm{commens}}:V_{\mathrm{syn}}\to V_{\mathrm{bio}}.$$

It only says that the multiplicity of an F-type V inside $\operatorname{Res}^{G_2}_F W'$ equals the multiplicity of W' inside the induced representation. It is a multiplicity-counting and adjunction statement, not a canonical cross-map between two different F-submodules.

In particular:

$$\operatorname{Hom}_{F}(L_2,U_6)=0,$$

because L_2 and U_6 are non-isomorphic irreducible real F-modules.

So if

$$V_{\mathrm{syn}}=L_2,\qquad V_{\mathrm{bio}}=U_6,$$

then there is no F_{21}-equivariant isomorphism between them, hence no G_2-equivariant one either.

Even if both occur inside the same restricted G_2-module

$$\operatorname{Res}^{G_2}_{F}W',$$

that only says they coexist as different F-isotypic blocks. It does not identify them.

Thus the answer to Q4b is:

$$\boxed{\text{Frobenius reciprocity is insufficient for commensurability.}}$$

The strict condition is:

$$\boxed{I_{\mathrm{syn}}\cong I_{\mathrm{bio}}\quad\text{as }F_{21}\text{-modules for }F_{21}\text{-level matching,}}$$

and

$$\boxed{I_{\mathrm{syn}}\cong I_{\mathrm{bio}}\quad\text{as }G_2\text{-modules for canonical }G_2\text{-commensurability.}}$$

A common G_2-envelope W' is not enough. The two scar image spaces must be the **same representation type**, not merely subrepresentations of the same ambient representation.

Q4c — Minimal sufficient conditions for μ_k transfer

Let

$$I_\bullet|_F\cong\mathbf 1^{\oplus m_1^\bullet}\oplus L_2^{\oplus m_2^\bullet}\oplus U_6^{\oplus m_6^\bullet}.$$

The F_{21}-multiplicity profile of a scar span is

$$\mu_k^\bullet=\bigl(\mu_{\mathbf 1}^\bullet(k),\mu_{L_2}^\bullet(k),\mu_{U_6}^\bullet(k)\bigr).$$

A clean sufficient condition for transfer is:

$$\boxed{m_1^{\mathrm{syn}}=m_1^{\mathrm{bio}},\qquad m_2^{\mathrm{syn}}=m_2^{\mathrm{bio}},\qquad m_6^{\mathrm{syn}}=m_6^{\mathrm{bio}},}$$

plus the existence of an F-equivariant isomorphism

$$\phi_F:I_{\mathrm{syn}}\to I_{\mathrm{bio}}$$

such that for every event e,

$$\boxed{\mathcal S_{\mathrm{bio}}(e)=\phi_F\bigl(\mathcal S_{\mathrm{syn}}(e)\bigr).}$$

Then for every event sequence $(e_1,\dots,e_k)$,

$$S_k^{\mathrm{bio}}=\phi_F(S_k^{\mathrm{syn}}),$$

$$\mathfrak S_k^{\mathrm{bio}}\cong\mathfrak S_k^{\mathrm{syn}},$$

and

$$\boxed{\mu_k^{\mathrm{bio}}=\mu_k^{\mathrm{syn}}}$$

componentwise.

The isotypic projectors make this explicit. For each real F-type

$$\tau\in\{\mathbf 1,L_2,U_6\},$$

let

$$\Pi_\tau^\bullet:I_\bullet\to I_{\bullet,\tau}$$

be the F-isotypic projector. If ϕ_F is F-equivariant, then

$$\boxed{\phi_F\Pi_\tau^{\mathrm{syn}}=\Pi_\tau^{\mathrm{bio}}\phi_F.}$$

Therefore

$$\dim \Pi_\tau^{\mathrm{syn}}S_k^{\mathrm{syn}}=\dim \Pi_\tau^{\mathrm{bio}}S_k^{\mathrm{bio}},$$

so the multiplicity profile transfers.

For full commensurability, replace ϕ_F by a G_2-equivariant isomorphism

$$\boxed{\phi_{G_2}:I_{\mathrm{syn}}\to I_{\mathrm{bio}}.}$$

A minimal strong sufficient condition is:

$$\boxed{I_{\mathrm{syn}}\cong U_\lambda,\qquad I_{\mathrm{bio}}\cong U_\lambda,}$$

where U_λ is the same real absolutely irreducible G_2-module, with multiplicity one.

Then

$$\operatorname{Hom}_{G_2}(I_{\mathrm{syn}},I_{\mathrm{bio}})\cong\mathbb R.$$

After fixing the invariant metric and orientation/sign convention,

$$\boxed{\phi_{\mathrm{commens}}\text{ is unique.}}$$

If instead

$$I_{\mathrm{syn}}\cong U_\lambda^{\oplus m},\qquad I_{\mathrm{bio}}\cong U_\lambda^{\oplus m},\qquad m>1,$$

then

$$\operatorname{End}_{G_2}(U_\lambda^{\oplus m})\cong M_m(\mathbb R)$$

for the adjoint-type real modules used here, so even full G_2 no longer gives uniqueness without an additional basis, metric, or event-ordering convention.

On “up to permutation of F_{21}-isotypic blocks”

For the real F_{21}-profile

$$(\mathbf 1,L_2,U_6),$$

there is no natural nontrivial permutation of the blocks:

$$\dim \mathbf 1=1,\qquad\dim L_2=2,\qquad\dim U_6=6.$$

So a block permutation cannot identify L_2 with U_6. At most, one may exchange conjugate complex constituents **inside** L_2 or inside U_6, but the real profile has already quotiented out that conjugation.

Thus the transfer condition is componentwise:

$$\boxed{\mu_{\mathbf 1}^{\mathrm{syn}}=\mu_{\mathbf 1}^{\mathrm{bio}},\qquad\mu_{L_2}^{\mathrm{syn}}=\mu_{L_2}^{\mathrm{bio}},\qquad\mu_{U_6}^{\mathrm{syn}}=\mu_{U_6}^{\mathrm{bio}}.}$$

Not merely “equal up to arbitrary permutation.”

Final theorem-style statement for Paper 12

Proposition Q4.

Let

$$I_{\mathrm{syn}}\subseteq \operatorname{End}(V_{\mathrm{syn}}),\qquad I_{\mathrm{bio}}\subseteq \operatorname{End}(V_{\mathrm{bio}})$$

be the real spans of the scar-encoding images, with F_{21} or G_2 acting by conjugation.

There exists a G_2-equivariant commensurability isomorphism

$$\phi_{\mathrm{commens}}:I_{\mathrm{syn}}\to I_{\mathrm{bio}}$$

aligning

$$\mathfrak J_k^{\mathrm{syn}}=\mathfrak J_k^{\mathrm{bio}}$$

for every event sequence if the following hold:

$$I_{\mathrm{syn}}\cong I_{\mathrm{bio}}\quad\text{as }G_2\text{-modules},$$

$$\mathcal S_{\mathrm{bio}}(e)=\phi_{\mathrm{commens}}\mathcal S_{\mathrm{syn}}(e)\quad\forall e,$$

and the normalization of ϕ_{commens} is fixed by the invariant metric and orientation/sign convention.

If both image spaces are one copy of the same absolutely irreducible G_2-module, then Schur gives

$$\operatorname{Hom}_{G_2}(I_{\mathrm{syn}},I_{\mathrm{bio}})\cong\mathbb R,$$

so the normalized ϕ_{commens} is unique.

By contrast, F_{21}-equivariance alone gives

$$\operatorname{End}_{F_{21}}(V)\cong M_{m_1}(\mathbb R)\oplus M_{m_2}(\mathbb C)\oplus M_{m_6}(\mathbb C),$$

so it leaves phase and multiplicity freedoms. Therefore F_{21} alone is not enough to fix a canonical commensurability map.

The usable hierarchy is:

$$\boxed{G_2\text{-commensurability}\Rightarrow F_{21}\text{-profile matching}\Rightarrow\mu_k^{\mathrm{syn}}=\mu_k^{\mathrm{bio}}.}$$

But the converse implications fail.

So the decision is:

$$\boxed{\text{Choose (ii): }F_{21}\text{ is insufficient; full }G_2\text{ commensurability is required.}}$$

For Paper 12, I would phrase P3 as:

$$\boxed{\text{P3 should test }F_{21}\text{-profile agreement as a necessary screen,}\text{ but the formal thesis requires }G_2\text{-commensurability.}}$$

Decision pinned. F_{21} is insufficient; full G_2 is required. The hierarchy $G_2 \to F_{21}$-profile $\to \mu_k$ transfer (no converses) is the Paper 12 §4 skeleton. P3 upgrades to a two-level test: F_{21}-profile screening as the necessary falsifier, G_2-commensurability as the sufficient canonical test. $\mathrm{End}_F(W) = M_2(\mathbb{C}) \oplus M_4(\mathbb{C})$ vs $\mathrm{End}_{G_2}(W) = M_2(\mathbb{R})$ is the quantitative gap.

Q5 — the end-to-end audit.

Take $\kappa \to 0$. The augmented flow becomes the unperturbed gradient flow:

$$\dot\theta = -\eta\, \nabla_\theta\, \mathcal{F}(\hat z(\theta)), \qquad \mathcal{F} = -\mathscr{C}_{\min}$$

with $\mathscr{C}_{\min}(\theta) = \min(\mathscr{C}_A(\theta), \mathscr{C}_B(\theta))$ and $\hat z(\theta) = M(\theta)^{-1}b$ the Paper 9 joint fixed point. Per Q1, treat $\mathcal{F}$ as branchwise C^ω with the analytic stratification $\Sigma_{\min} \cup \Sigma_\Theta \cup Z$ (here Σ_Θ and Z are inactive at $\kappa = 0$; only $\Sigma_{\min}$ remains).

Paper 9's 50-seed Monte Carlo (Appendix V.3.i; CSV is computations/paper9_conjecture95_mc_seeds.csv) reports for each seed pair (s_a, s_b) = (20260504+k, 20260505+k), k = 0,...,49:

- The optimal coupling angle θ★ at $\varphi = 0$ (aligned bias, b_A = e_1, b_B = e_1) ranges [0°, 75.8°] across the 50 seeds with mean 3.2°.
- All 50 seeds satisfy $|\theta★ - 45°| > 5°$.
- The seeded geometry of Proposition 9.5 (the original verified seed pair, seeds 20260504 and 20260505 specifically — k=0) has θ★ ≈ 51° at the highest end of the range.

Setup. Define $c(\theta) := \mathscr{C}\_\min(\hat{z}(\theta))$ at $\varphi = 0$ for a given seed pair (R_A, R_B). The κ = 0 gradient flow has critical points exactly where $c'(\theta) = 0$ (equivalently, where $\nabla\_\theta \mathcal{F} = 0$).

The question. Is each Paper 9 MC θ★_seed a critical point of c(θ) on [0, π/2]? I.e., for every seed k = 0,...,49, does

$c'(\theta★\_k) = 0$?

This question is binary. Either Paper 12's κ = 0 limit recovers Paper 9's MC observations as gradient-flow attractors, or it doesn't. Both answers are publishable, but they have different consequences:

- If yes: Paper 12 is consistent with Paper 9, and the augmented flow is the right nonlinear extension. Proceed to draft §3.
- If no: either $\mathcal{F} = -\mathscr{C}\_\min$ is the wrong functional (and we need to pick another that has the MC θ★ values as its critical points), or Paper 9's Proposition 9.5(c) "θ★ = arg max $\mathscr{C}$_min(θ)" needs revisiting (and maybe the MC actually finds something other than gradient critical points). Either is a real result that has to be reconciled before §3 drafts.

Two specific sub-questions:

(Q5a) Specify exactly the numerical experiment Φ should run to verify. I want a paste-into-Python spec including:
- the function c(θ) computation (M(θ)^{-1}b, the cosine-similarity-min structure)
- the derivative c'(θ) computation (analytic via M'(θ) or numerical finite-difference; specify which)
- the MC θ★ values to check against (read from paper9_conjecture95_mc_seeds.csv)
- the agreement tolerance: what counts as |c'(θ★)| ≈ 0? Specify an absolute tolerance with units, not just

"small" — if $|c'(\theta★)| < 1e-6$ counts as agreement, say so explicitly
- the failure-mode classification: if some seeds agree and others don't, what's the cutoff for declaring the recovery "passes" vs "fails"?

Φ has Paper 9's exact code base, so referencing its functions (e.g., the 28×28 M(θ) construction in build_mc_seeds.py) is fine.

(Q5b) Predict the outcome. Based on the structure of Paper 9 ($\mathscr{C}$_min has piecewise-C^ω regularity per Q1; Proposition 9.5(c) defines θ★ as the arg max of $\mathscr{C}$_min(θ) at φ=0; 50/50 seeds satisfy |θ★ − 45°| > 5°), do you expect c'(θ★) = 0 to hold for all 50 seeds? Specifically:
- if θ★ is in the interior of the smooth branch $\mathscr{C}$_A or $\mathscr{C}$_B (not on Σ_min): yes, c'(θ★) = 0 is forced by arg-max-of-smooth-function
- if θ★ is on the boundary θ = 0 or θ = π/2: c'(θ★) may be nonzero (boundary maximum)
- if θ★ is on Σ_min = {$\mathscr{C}$_A = $\mathscr{C}$_B}: c is continuous but c' has a jump; the "critical point" condition needs sub-differential treatment

Tell me which case dominates the 50-seed ensemble. The answer for k=0 (seeds 20260504, 20260505, where θ★ ≈ 51°, comfortably interior): I predict c'(51°) = 0 cleanly. For seeds with θ★ near 0° (the bulk of the MC ensemble, mean 3.2°): boundary case, less clean.

This question is binary in conclusion but layered in mechanism. Don't hedge the conclusion; do hedge the mechanism per case.

Q5 conclusion

\boxed{\text{No: I do not expect all 50 Paper 9 MC }\theta^\star_k\text{ to satisfy }c'(\theta^\star_k)=0.}

I expect the audit to split into three classes:

\boxed{\text{interior smooth optima: }c'(\theta^\star)=0}

\boxed{\text{boundary optima near }0^\circ\text{: ordinary }c'(\theta^\star)=0\text{ generally fails, but KKT/projection passes}}

\boxed{\text{possible }\Sigma_{\min}\text{ optima: }c'\text{ is undefined, but Clarke/subgradient criticality may pass}}

So the strict binary statement

$$\forall k,\quad c'(\theta^\star_k)=0$$

will probably fail. The corrected $\kappa\to0$ recovery statement should be:

$$\boxed{\theta^\star_k\text{ is a stationary point of the constrained nonsmooth gradient system for }c=\min(c_A,c_B)\text{ on }[0,\pi/2].}$$

That means:

$$\boxed{\text{interior smooth: }c'=0;\qquad\text{boundary: one-sided KKT condition;}\qquad\Sigma_{\min}:\ 0\in\partial_C c.}$$

This is not fatal to Paper 12, but it does mean the $\kappa\to0$ audit should not be phrased as “all MC $\theta^\star$ are ordinary critical points.” It should be phrased as “all MC $\theta^\star$ are constrained/nonsmooth stationary argmax points.”

Q5a — Exact numerical experiment for Φ

Use the analytic derivative as the primary test.

The derivative units are:

$$\boxed{\text{dimensionless coherence per radian}.}$$

The strict agreement tolerance should be:

$$\boxed{|c'(\theta^\star)|<10^{-6}\quad\text{coherence/radian}.}$$

Use finite differences only as a sanity check, not as the primary derivative, because $c=\min(c_A,c_B)$ is nondifferentiable on $\Sigma_{\min}$.

Mathematical computation

At $\varphi=0$,

$$b_A=e_1,\qquad b_B=e_1,$$

and

$$b=\begin{pmatrix}b_A\\b_B\end{pmatrix}.$$

Use the Paper 9 fixed point

$$\hat z(\theta)=M(\theta)^{-1}b.$$

With the convention

$$M(\theta)=\Psi_{-\theta}-\operatorname{diag}(r_A R_A,r_B R_B),$$

one has

$$M(\theta)=\begin{pmatrix}\cos\theta\,I-r_A R_A & \sin\theta\,I\\-\sin\theta\,I & \cos\theta\,I-r_B R_B\end{pmatrix},$$

and

$$M'(\theta)=\begin{pmatrix}-\sin\theta\,I & \cos\theta\,I\\-\cos\theta\,I & -\sin\theta\,I\end{pmatrix}.$$

Then

$$\hat z'(\theta)=-M(\theta)^{-1}M'(\theta)\hat z(\theta).$$

Write

$$\hat z(\theta)=(\hat x(\theta),\hat y(\theta)),\qquad\hat z'(\theta)=(\hat x'(\theta),\hat y'(\theta)).$$

The two branch coherences are

$$c_A(\theta)=\frac{\langle \hat x(\theta),b_A\rangle}{\|\hat x(\theta)\|\,\|b_A\|},$$

$$c_B(\theta)=\frac{\langle \hat y(\theta),b_B\rangle}{\|\hat y(\theta)\|\,\|b_B\|}.$$

Then

$$c(\theta)=\min(c_A(\theta),c_B(\theta)).$$

For a branch

$$h(u)=\frac{\langle u,b\rangle}{\|u\|\,\|b\|},$$

the derivative along u' is

$$Dh[u']=\frac{\langle u',b\rangle}{\|u\|\,\|b\|}-\frac{\langle u,b\rangle\langle u,u'\rangle}{\|u\|^3\,\|b\|}.$$

So

$$c_A'(\theta)=Dh_{\hat x(\theta)}[\hat x'(\theta)],$$

$$c_B'(\theta)=Dh_{\hat y(\theta)}[\hat y'(\theta)].$$

Away from

$$\Sigma_{\min}=\{c_A=c_B\},$$

the derivative of c is

$$c'(\theta)=\begin{cases}c_A'(\theta),&c_A(\theta)<c_B(\theta),\\c_B'(\theta),&c_B(\theta)<c_A(\theta).\end{cases}$$

At

$$c_A=c_B,$$

ordinary c' is undefined unless the two branch derivatives agree. The Clarke criticality check is

$$\boxed{0\in \operatorname{conv}\{c_A'(\theta),c_B'(\theta)\}.}$$

In one dimension, this is

$$\boxed{\min(c_A',c_B')\le 0\le \max(c_A',c_B').}$$

Paste-into-Python audit script

This is written so \Phi only has to bind the Paper 9 seed/geometry constructor if the names differ.

```
import numpy as np

import pandas as pd

from pathlib import Path

from numpy.linalg import solve, norm, cond


PI2 = 0.5 * np.pi


# -----------------------------
```

```
# Audit tolerances
# -----------------------------

DERIV_ATOL = 1e-6         # coherence per radian
MIN_TIE_ATOL = 1e-8       # coherence units
BOUNDARY_ATOL = 1e-10     # radians
FD_H = 1e-5               # radians, finite-difference diagnostic only
NEAR_BOUNDARY_DIAG = np.deg2rad(0.25)

CSV_PATH = Path("computations/paper9_conjecture95_mc_seeds.csv")


# -----------------------------
# Paper 9 adapter
# -----------------------------
# Replace this adapter with the exact Paper 9 helper if its function
# names differ. The expected return values are:
#   R_A, R_B : 14x14 orthogonal/representation matrices
#   r_A, r_B : scalar contraction rates
#   b_A, b_B : 14-vectors, at phi=0 usually e_1, e_1
#
# If build_mc_seeds.py already exposes M(theta) directly, use that
# exact M(theta) and Mprime(theta) instead of the formula below.
```

```
def load_seed_geometry(s_a: int, s_b: int):
    """
    Paper 9 repo adapter.

    Expected behavior:
      - reconstruct the exact seed geometry used in Appendix V.3.i
      - force phi=0 / aligned bias
      - return R_A, R_B, r_A, r_B, b_A, b_B
    """
    try:
        # Example name; adjust to the actual Paper 9 codebase if needed.
        from computations.build_mc_seeds import build_seed_pair_geometry
        return build_seed_pair_geometry(s_a=s_a, s_b=s_b, phi=0.0)
    except ImportError as e:
        raise ImportError(
            "Bind load_seed_geometry() to the exact Paper 9 seed constructor "
            "in computations/build_mc_seeds.py."
        ) from e


# -----------------------------
# M(theta), zhat(theta), derivative
# -----------------------------
```

```
def M_and_Mprime(theta, R_A, R_B, r_A, r_B):
    """
    Convention consistent with:
        zhat(theta) = M(theta)^(-1) (b_A; b_B)
    and
        M(theta) = Psi_{-theta} - diag(r_A R_A, r_B R_B).

    If Paper 9's build_mc_seeds.py has an exact M(theta) function,
    use that exact function and its differentiated Mprime.
    """
    I = np.eye(14)
    c = np.cos(theta)
    s = np.sin(theta)

    M = np.block([
        [c * I - r_A * R_A,   s * I],
        [-s * I,              c * I - r_B * R_B],
    ])

    Mp = np.block([
        [-s * I,   c * I],
        [-c * I,  -s * I],
    ])
```

```
    return M, Mp


def zhat_and_zprime(theta, geom):
    R_A, R_B, r_A, r_B, b_A, b_B = geom
    b = np.concatenate([b_A, b_B])

    M, Mp = M_and_Mprime(theta, R_A, R_B, r_A, r_B)

    z = solve(M, b)
    zp = -solve(M, Mp @ z)

    return z, zp, cond(M)


# -----------------------------
# Coherence and analytic branch derivatives
# -----------------------------

def cos_sim(u, b):
    nu = norm(u)
    nb = norm(b)
    if nu == 0.0 or nb == 0.0:
        return np.nan
```

```
    return float(np.dot(u, b) / (nu * nb))



def cos_sim_derivative(u, up, b):
    nu = norm(u)
    nb = norm(b)
    if nu == 0.0 or nb == 0.0:
        return np.nan

    term1 = np.dot(up, b) / (nu * nb)
    term2 = (np.dot(u, b) * np.dot(u, up)) / (nu**3 * nb)
    return float(term1 - term2)



def coherence_components(theta, geom):
    R_A, R_B, r_A, r_B, b_A, b_B = geom

    z, zp, cond_M = zhat_and_zprime(theta, geom)

    x = z[:14]
    y = z[14:]
    xp = zp[:14]
    yp = zp[14:]
```

```
    c_A = cos_sim(x, b_A)
    c_B = cos_sim(y, b_B)

    dc_A = cos_sim_derivative(x, xp, b_A)
    dc_B = cos_sim_derivative(y, yp, b_B)

    return {
        "c_A": c_A,
        "c_B": c_B,
        "dc_A": dc_A,
        "dc_B": dc_B,
        "c_min": min(c_A, c_B),
        "cond_M": cond_M,
    }


def c_min_scalar(theta, geom):
    q = coherence_components(theta, geom)
    return q["c_min"]


def active_branch_derivative(theta, geom):
    q = coherence_components(theta, geom)
    c_A, c_B = q["c_A"], q["c_B"]
```

```
    dc_A, dc_B = q["dc_A"], q["dc_B"]

    if c_A < c_B - MIN_TIE_ATOL:
        return dc_A, "A", q

    if c_B < c_A - MIN_TIE_ATOL:
        return dc_B, "B", q

    # On Sigma_min: ordinary derivative is not defined unless dc_A ~= dc_B.
    if abs(dc_A - dc_B) < DERIV_ATOL:
        return 0.5 * (dc_A + dc_B), "Sigma_min_equal_slope", q

    return np.nan, "Sigma_min_kink", q


# -----------------------------
# Finite-difference diagnostics
# -----------------------------

def one_sided_slopes(theta, geom, h=FD_H):
    c0 = c_min_scalar(theta, geom)

    left = np.nan
    right = np.nan
```

```
    if theta - h >= 0.0:
        left = (c0 - c_min_scalar(theta - h, geom)) / h

    if theta + h <= PI2:
        right = (c_min_scalar(theta + h, geom) - c0) / h

    return left, right


def centered_fd_derivative(theta, geom, h=FD_H):
    if theta - h < 0.0 or theta + h > PI2:
        return np.nan
    return (c_min_scalar(theta + h, geom) - c_min_scalar(theta - h, geom)) / (2.0 * h)


# -----------------------------
# CSV helpers
# -----------------------------

def get_col(row, candidates):
    for c in candidates:
        if c in row.index:
            return row[c]
```

```
    return None


def get_theta_star(row):
    rad = get_col(row, [
        "theta_star_rad",
        "theta_opt_rad",
        "theta_rad",
        "theta_best_rad",
    ])

    if rad is not None and not pd.isna(rad):
        return float(rad)

    deg = get_col(row, [
        "theta_star_deg",
        "theta_opt_deg",
        "theta_deg",
        "theta_best_deg",
    ])

    if deg is not None and not pd.isna(deg):
        return float(np.deg2rad(deg))
```

```
    raise KeyError(
        "Could not find theta-star column. Expected one of "
        "theta_star_rad/theta_opt_rad/theta_rad/theta_best_rad "
        "or theta_star_deg/theta_opt_deg/theta_deg/theta_best_deg."
    )


def get_seeds(row, k):
    s_a = get_col(row, ["s_a", "seed_a", "seed_A"])
    s_b = get_col(row, ["s_b", "seed_b", "seed_B"])

    if s_a is None or pd.isna(s_a):
        s_a = 20260504 + k

    if s_b is None or pd.isna(s_b):
        s_b = 20260505 + k

    return int(s_a), int(s_b)


# -----------------------------
# Classification logic
# -----------------------------
```

```
def classify_theta_star(theta, geom):
    d_active, branch, q = active_branch_derivative(theta, geom)
    left_slope, right_slope = one_sided_slopes(theta, geom)
    fd_center = centered_fd_derivative(theta, geom)

    c_A, c_B = q["c_A"], q["c_B"]
    dc_A, dc_B = q["dc_A"], q["dc_B"]

    on_lower_boundary = theta <= BOUNDARY_ATOL
    on_upper_boundary = theta >= PI2 - BOUNDARY_ATOL
    near_lower_boundary = theta <= NEAR_BOUNDARY_DIAG
    near_upper_boundary = theta >= PI2 - NEAR_BOUNDARY_DIAG
    on_sigma_min = abs(c_A - c_B) <= MIN_TIE_ATOL

    # Strict ordinary-critical test:
    #   pass only when c' exists and |c'| < DERIV_ATOL.
    strict_cprime_zero = False

    if branch in ["A", "B", "Sigma_min_equal_slope"]:
        strict_cprime_zero = abs(d_active) < DERIV_ATOL

    # Clarke/subgradient condition for Sigma_min.
    clarke_zero = False
    if on_sigma_min:
```

```
    lo = min(dc_A, dc_B)
    hi = max(dc_A, dc_B)
    clarke_zero = (lo <= DERIV_ATOL) and (hi >= -DERIV_ATOL)

# One-sided KKT condition for maximizing c on [0, pi/2].
#
# Lower boundary maximum:
#   moving right should not increase c, so right_slope <= 0.
#
# Upper boundary maximum:
#   moving into the boundary from the left should not decrease into the boundary;
#   equivalently left_slope >= 0.
#
# Interior smooth maximum:
#   derivative zero.
#
# Interior kink maximum:
#   left_slope >= 0 and right_slope <= 0.
kkt_stationary = False
kkt_reason = "unclassified"

if on_lower_boundary:
    kkt_stationary = (not np.isnan(right_slope)) and (right_slope <= DERIV_ATOL)
    kkt_reason = "lower_boundary_KKT"
```

```
elif on_upper_boundary:
    kkt_stationary = (not np.isnan(left_slope)) and (left_slope >= -DERIV_ATOL)
    kkt_reason = "upper_boundary_KKT"

elif branch in ["A", "B", "Sigma_min_equal_slope"]:
    kkt_stationary = strict_cprime_zero
    kkt_reason = f"smooth_or_equal_slope_{branch}"

elif branch == "Sigma_min_kink":
    left_ok = (not np.isnan(left_slope)) and (left_slope >= -DERIV_ATOL)
    right_ok = (not np.isnan(right_slope)) and (right_slope <= DERIV_ATOL)
    kkt_stationary = clarke_zero and left_ok and right_ok
    kkt_reason = "Sigma_min_Clarke_KKT"

# Severity classification.
if strict_cprime_zero:
    audit_class = "STRICT_PASS_cprime_zero"

elif kkt_stationary:
    if on_lower_boundary or on_upper_boundary:
        audit_class = "KKT_ONLY_boundary_not_ordinary_cprime"
    elif branch == "Sigma_min_kink":
        audit_class = "KKT_ONLY_Sigma_min_kink_not_ordinary_cprime"
```

```
    else:
        audit_class = "KKT_ONLY_other"

else:
    audit_class = "FAIL_not_stationary_for_cmin"

return {
    "theta_rad": theta,
    "theta_deg": float(np.rad2deg(theta)),
    "branch": branch,
    "c_A": c_A,
    "c_B": c_B,
    "c_A_minus_c_B": c_A - c_B,
    "dc_A": dc_A,
    "dc_B": dc_B,
    "d_active": d_active,
    "fd_center": fd_center,
    "left_slope": left_slope,
    "right_slope": right_slope,
    "on_sigma_min": on_sigma_min,
    "on_lower_boundary": on_lower_boundary,
    "on_upper_boundary": on_upper_boundary,
    "near_lower_boundary_diag": near_lower_boundary,
    "near_upper_boundary_diag": near_upper_boundary,
```

```
        "clarke_zero": clarke_zero,
        "strict_cprime_zero": strict_cprime_zero,
        "kkt_stationary": kkt_stationary,
        "kkt_reason": kkt_reason,
        "audit_class": audit_class,
        "cond_M": q["cond_M"],
    }


# -----------------------------
# Main audit
# -----------------------------

def run_q5_audit():
    df = pd.read_csv(CSV_PATH)

    rows = []

    for k, row in df.iterrows():
        s_a, s_b = get_seeds(row, k)
        theta = get_theta_star(row)

        geom = load_seed_geometry(s_a, s_b)
        result = classify_theta_star(theta, geom)
```

```
    result.update({
        "k": int(k),
        "s_a": s_a,
        "s_b": s_b,
    })

    rows.append(result)

out = pd.DataFrame(rows)

n = len(out)
n_strict = int(out["strict_cprime_zero"].sum())
n_kkt = int(out["kkt_stationary"].sum())
n_fail = int((~out["kkt_stationary"]).sum())

print("Q5 kappa->0 audit")
print("=================")
print(f"n seeds:                  {n}")
print(f"strict cprime-zero pass: {n_strict}/{n}")
print(f"KKT/Clarke pass:         {n_kkt}/{n}")
print(f"KKT/Clarke fail:         {n_fail}/{n}")
print()
print(out["audit_class"].value_counts(dropna=False))
```

```
    print()
    print("Theta-star range deg:", out["theta_deg"].min(), out["theta_deg"].max())
    print("Theta-star mean deg: ", out["theta_deg"].mean())
    print("Max |active derivative| among strict-defined rows:",
          np.nanmax(np.abs(out["d_active"].to_numpy(dtype=float))))

    out.to_csv("paper12_q5_kappa0_criticality_audit.csv", index=False)

    # Binary strict test required by Q5:
    strict_recovery_passes = (n_strict == n)

    # Corrected constrained/nonsmooth recovery:
    constrained_recovery_passes = (n_kkt == n)

    print()
    print(f"STRICT_RECOVERY_PASSES = {strict_recovery_passes}")
    print(f"CONSTRAINED_NONSmooth_RECOVERY_PASSES =
{constrained_recovery_passes}")

    return out


if __name__ == "__main__":
    audit = run_q5_audit()
```

Optional local re-optimization diagnostic

The primary audit above checks the \theta^\star values recorded in the CSV. If the CSV stores rounded angles, derivative tests can falsely fail. So \Phi should optionally rerun a high-precision local bounded optimization and compare.

```
from scipy.optimize import minimize_scalar


def reoptimize_theta_star(geom):

    res = minimize_scalar(

        lambda th: -c_min_scalar(th, geom),

        bounds=(0.0, PI2),

        method="bounded",

        options={"xatol": 1e-12, "maxiter": 1000},

    )

    return float(res.x), float(-res.fun), res


def compare_csv_to_reopt(row, k):

    s_a, s_b = get_seeds(row, k)

    theta_csv = get_theta_star(row)

    geom = load_seed_geometry(s_a, s_b)


    theta_opt, c_opt, res = reoptimize_theta_star(geom)


    return {

        "k": k,

        "s_a": s_a,
```

```
    "s_b": s_b,
    "theta_csv_deg": np.rad2deg(theta_csv),
    "theta_reopt_deg": np.rad2deg(theta_opt),
    "abs_delta_theta_rad": abs(theta_csv - theta_opt),
    "abs_delta_theta_deg": abs(np.rad2deg(theta_csv - theta_opt)),
    "c_reopt": c_opt,
    "success": bool(res.success),
  }
```

Recommended angle agreement diagnostic:

$$\boxed{|\theta^\star_{\mathrm{CSV}}-\theta^\star_{\mathrm{reopt}}|<10^{-5}\ \mathrm{rad}}$$

for unrounded CSV values.

If the CSV stores rounded degrees, use the derivative test on the reoptimized $\theta^\star_{\mathrm{reopt}}$, not on the rounded value.

Pass/fail rule

Use two separate pass/fail labels.

Strict Q5 test

This is the literal question:

$$\forall k,\quad c'(\theta^\star_k)=0?$$

Declare:

$$\boxed{\text{STRICT PASS}\iff50/50\text{ seeds satisfy }|c'(\theta^\star_k)|<10^{-6}.}$$

Anything else is:

$$\boxed{\text{STRICT FAIL.}}$$

No majority rule. If 49/50 pass, the universal statement fails.

Corrected κ→0 recovery test

Declare:

$$\boxed{\text{CONSTRAINED/NONSMOOTH PASS}}$$

iff all 50 seeds satisfy one of:

1. Interior smooth branch:

$$|c'(\theta^\star_k)|<10^{-6}.$$

2. Lower boundary:

$$\theta^\star_k=0,\qquad c'_+(0)\le 10^{-6}.$$

3. Upper boundary:

$$\theta^\star_k=\pi/2,\qquad c'_-(\pi/2)\ge -10^{-6}.$$

4. Min-kink:

$$\theta^\star_k\in\Sigma_{\min},\qquad 0\in \operatorname{conv}\{c_A'(\theta^\star_k),c_B'(\theta^\star_k)\},$$

and the one-sided slopes satisfy the local maximum condition

$$c'_-(\theta^\star_k)\ge -10^{-6},\qquad c'_+(\theta^\star_k)\le 10^{-6}.$$

If any seed is interior smooth with

$$|c'(\theta^\star_k)|\ge 10^{-6}$$

and is not boundary or $\Sigma_{\min}$-critical, then this is a genuine failure:

$$\boxed{\text{either }\mathcal F=-\mathcal C_{\min}\text{ is the wrong κ→0 functional,}\text{ or Paper 9's }\theta^\star\text{ is not the argmax of }c.}$$

Q5b — Predicted outcome

My prediction is:

$$\boxed{\text{STRICT FAIL: not all 50 satisfy ordinary }c'(\theta^\star)=0.}$$

But also:

$$\boxed{\text{CONSTRAINED/NONSMOOTH PASS is likely.}}$$

That is, I expect the Paper 9 MC optima to satisfy the correct first-order conditions for maximizing

$$c(\theta)=\min(c_A(\theta),c_B(\theta))$$

on the compact interval

$$[0,\pi/2],$$

but not all to satisfy the ordinary interior equation

$$c'(\theta)=0.$$

Expected class distribution

The ensemble is dominated by the lower-boundary case.

The reported range and mean,

$$\theta^\star\in[0^\circ,75.8^\circ],\qquad\overline{\theta^\star}=3.2^\circ,$$

with one known high-end seed around

$$51^\circ,$$

strongly implies that most of the 50 seeds are at or very near

$$\theta^\star=0^\circ.$$

For those seeds, the correct maximum condition is not

$$c'(0)=0.$$

It is the one-sided KKT condition

$$\boxed{c'_+(0)\le0.}$$

Equivalently, for the gradient flow

$$\dot\theta=\eta c'(\theta)$$

on the constrained interval, the vector field at the boundary points outward or tangentially, and the projected flow stays at

$$\theta=0.$$

Thus these seeds are Paper 9-consistent but fail the literal ordinary-critical-point test unless an accidental derivative cancellation occurs.

Expected outcome for $k=0$

For the original Proposition 9.5 seed pair,

$$(s_a,s_b)=(20260504,20260505),$$

with

$$\theta^\star\approx 51^\circ,$$

I expect an interior pass if it lies off $\Sigma_{\min}$:

$$\boxed{|c'(51^\circ)|<10^{-6}.}$$

That seed is comfortably interior, so if it is on a smooth active branch, the argmax condition forces

$$c'(\theta^\star)=0.$$

One caveat: interior does not automatically mean smooth. If the optimum occurs exactly on

$$\Sigma_{\min}=\{c_A=c_B\},$$

then ordinary c' is undefined and the correct condition is instead

$$0\in \operatorname{conv}\{c_A',c_B'\}.$$

I would still expect $k=0$ to pass the corrected constrained/nonsmooth audit.

Consequence for Paper 12

The $\kappa\to 0$ limit should be stated as a projected/Clarke gradient system:

$$\boxed{\dot\theta\in\Pi_{T_{[0,\pi/2]}(\theta)}\bigl(\eta\,\partial_C c(\theta)\bigr),\qquad c=\mathcal C_{\min}.}$$

On smooth interior branches this reduces to

$$\dot\theta=\eta c'(\theta).$$

At $\Sigma_{\min}$, use the Clarke subdifferential. At the endpoints, use the tangent cone of

$$[0,\pi/2].$$

So the audit answer is binary but not destructive:

$$\boxed{\text{Ordinary critical-point recovery: predicted NO.}}$$

$$\boxed{\text{Constrained nonsmooth argmax recovery: predicted YES.}}$$

If Φ finds an interior smooth seed with nonzero derivative above

$$10^{-6}$$

after reoptimizing $\theta^\star$, then that is the real red flag. Boundary-only failures merely mean Paper 12 §3 must use the constrained nonsmooth gradient formulation rather than the oversimplified condition $c'=0$ everywhere.

Last question — meta.

You've given me answers to Q2, Q1, Q3, Q4, Q5 in that order. Rank them by your *own* confidence:

1. Most confident answer (you'd defend it against a hostile reviewer).
2. ...
3. ...
4. ...
5. Least confident answer (the one most likely to be wrong).

For the least confident answer, identify:
(a) The single load-bearing assumption that, if wrong, would invalidate the answer.
(b) The cheapest experiment (numerical or analytical) that could falsify the assumption.
(c) Whether you'd recommend I have Φ verify it, get a second opinion from another model, or punt the question to a subsequent paper.

For the *most* confident answer, identify the one place where you think mainstream mathematics literature would push back, even if you think the pushback is wrong.

Don't be polite. I'd rather hear "I'm 30% confident in Q4 and here's why" than "all five answers are reasonable starting points."

Confidence ranking

1.

Q2 — Schur-derived $\xi_\star$

Confidence: ~92%.

I would defend this one hardest.

The core claim is standard:

$W=V^{14}\oplus V^{14}\cong V^{14}\otimes \mathbb R^2$,

with G_2 acting irreducibly on V^{14} and trivially on the multiplicity factor. Therefore

$\operatorname{End}_{G_2}(W)\cong M_2(\mathbb R)$.

Imposing the product metric gives

$O(2)$,

and choosing the connected orientation-preserving branch gives

$SO(2)$.

Hence the infinitesimal Schur direction is the unique line

$\mathbb R\,\partial_\theta$

with generator

$$
\mathscr J=
\begin{pmatrix}
0&-I_{14}\\
I_{14}&0
\end{pmatrix}.
$$

The only extra modeling step is the sign rule

$$\xi_\star=\operatorname{sgn}(c'(\theta))\,\partial_\theta.$$

That part is not Schur-derived; it is coherence-objective-derived. I stated that distinction, so I’m comfortable with the answer.

Mainstream pushback I would expect:
A reviewer may say:

$$\text{“Equivariance + isometry gives }O(2),\text{ not a canonical oriented }SO(2)\text{ generator.”}$$

That pushback is technically fair. The Schur argument gives the O(2) fiber; choosing SO(2), choosing orientation, and then choosing the sign by $c'(\theta)$ are additional conventions. My response would be: yes, but once Paper 9 has already fixed the oriented Schur circle Ψ_θ, Paper 12 inherits that orientation, and then the only remaining ambiguity is the sign, which $\mathcal C_{\min}$ fixes except on $Z=\{c'=0\}$.

So Q2 is solid, provided the Paper 9 orientation convention is accepted.

2.

Q1 — piecewise-real-analytic regularity / Σ_{tot}

Confidence: ~84%.

This is also strong.

The claim was:

$$\mathcal F=-\mathcal C_{\min}$$

should be treated as branchwise C^ω, not globally analytic, because

$$\mathcal C_{\min}=\min(\mathcal C_A,\mathcal C_B)$$

is kinked on

$$\Sigma_{\min}=\{\mathcal C_A=\mathcal C_B\}.$$

The event stratification

$$\Sigma_{\mathrm{tot}}=\Sigma_{\min}\cup\Sigma_\Theta\cup Z\cup\mathcal G_{\mathrm{NP}}$$

is the right structural object. Transverse crossings are Carathéodory, sliding on $\Sigma_{\min}$ is Filippov, and actual NP tears are hybrid jumps.

I’m confident because this is the standard nonsmooth/hybrid systems decomposition.

The weak point is not the math. The weak point is whether Paper 12 wants Θ sharp or smoothed. I defended “sharp” because the model is explicitly about tears. A smoothing reviewer could reasonably say: “Then you are studying a discontinuous/hybrid limit, not a classical ODE.” That is true. But that is exactly the point.

3.

Q4 — F_{21} insufficient, full G_2 required

Confidence: ~74%.

I’m fairly confident in the conclusion:

$$\boxed{F_{21}\text{ alone is too weak to determine a canonical }\phi_{\mathrm{commens}}.}$$

The quantitative gap

$$\operatorname{End}_{F_{21}}(W)\cong M_2(\mathbb C)\oplus M_4(\mathbb C)$$

versus

$$\operatorname{End}_{G_2}(W)\cong M_2(\mathbb R)$$

is exactly the kind of commutant blow-up that kills uniqueness.

The part I’m less confident about is the exact biological/synthetic formalization:

$$\mathcal S_{\mathrm{syn}}:e\mapsto \operatorname{End}(V_{\mathrm{syn}}),\qquad\mathcal S_{\mathrm{bio}}:e\mapsto \operatorname{Proj}(V_{\mathrm{bio}}).$$

I treated the scar image spans as representation spaces under conjugation. That is mathematically natural, but it is still a modeling decision. If Paper 12 instead treats scars as vectors in carrier spaces, rather than operators/projectors in endomorphism spaces, the representation bookkeeping changes.

But the headline conclusion survives:

$$F_{21}\text{-profile matching is necessary but not sufficient for canonical commensurability.}$$

The other possible weak point is the exact restriction

$$V^{14}|_{F_{21}}\cong L_2\oplus 2U_6.$$

I believe it is right for the Fano-orientation embedding, but Φ should verify the character calculation from the exact matrices used in the archive.

4.

Q3 — scar invariant $\mathfrak J_k=(S_k,\mathfrak S_k,\mu_k)$

Confidence: ~61%.

This one is useful, but it is the most “framework-building” answer, not a theorem forced by existing literature.

I’m confident in the negative parts:

$$\text{winding number is not injective in scar count;}$$

$$\text{persistent }H_1\text{ is not injective in scar count;}$$

$$\dim S_k\le 28\text{ and therefore saturates.}$$

Those are solid.

The less certain part is calling

$$\mathfrak S_k=\bigoplus_{i=1}^k\mathbb R[F_{21}]v_i$$

the “canonical scar invariant.” It is count-faithful, computationally cheap, and representation-compatible. But it is not a topological invariant in the usual sense. It is a representation-theoretic event ledger.

So a hostile reviewer could say:

$$\text{“You were asked for a topological scar invariant and you returned a module-valued audit log.”}$$

That criticism is not wrong. My defense would be: for actual scar count, topology is the wrong tool; the event-indexed representation module is the invariant that records exactly what Paper 12 cares about. But the language should be tightened:

$$\boxed{\mathfrak J_k\text{ is a representation-theoretic scar invariant, not a pure topological invariant.}}$$

If §3 or §4 says “topological invariant” too strongly, I would soften it.

5.

Q5 — κ→0 audit prediction

Confidence: ~43%.

This is the least confident answer.

The numerical spec is good. The prediction is the weak part.

I predicted:

$$\boxed{\text{strict ordinary derivative-zero recovery fails for all 50.}}$$

That prediction rests on interpreting the reported MC mean and range as evidence that many seeds are true boundary optima near

$$\theta^\star=0.$$

But I did **not** execute Paper 9’s code or inspect the CSV. So this is an inference from summary statistics, not a verified result.

The audit spec is probably useful. The outcome prediction could easily be wrong.

Least confident answer: Q5

(a) Single load-bearing assumption

The load-bearing assumption is:

$$\boxed{\text{The many near-zero }\theta^\star\text{ values are true boundary/constrained optima, not rounded interior critical points.}}$$

If those near-zero values are actually small positive interior maximizers, or if the CSV reports rounded degrees while the true optimizer satisfies

$$c'(\theta^\star)=0,$$

then my “strict fail” prediction could be wrong.

In particular, if Paper 9’s optimizer internally found

$$\theta^\star_k>0$$

for most seeds but the CSV rounded them to 0°, then evaluating c' at the rounded CSV value would falsely suggest noncriticality. The right audit would need to reoptimize at high precision.

(b) Cheapest falsifying experiment

Run the Q5 derivative audit on the actual CSV and exact Paper 9 $M(\theta)$, but include local reoptimization.

For each seed:

1. Reconstruct R_A,R_B,b_A,b_B at $\varphi=0$.
2. Recompute

$$\theta^\star_{\mathrm{reopt}}=\operatorname*{arg\,max}_{\theta\in[0,\pi/2]}c(\theta)$$

with high precision.
3. Compute analytic

$$\hat z'(\theta)=-M(\theta)^{-1}M'(\theta)\hat z(\theta).$$

4. Compute active branch derivatives

$$c_A'(\theta),\qquad c_B'(\theta).$$

5. Classify each seed as:

$$\text{smooth interior } |c'|<10^{-6},$$

$$\text{boundary KKT},$$

or

$$\Sigma_{\min}\text{ Clarke critical}.$$

The single falsifier of my prediction is:

$$\boxed{50/50\text{ reoptimized }\theta^\star_k\text{ are smooth interior or equal-slope points with }|c'(\theta^\star_k)|<10^{-6}.}$$

If that happens, Q5's predicted “strict fail” is wrong.

(c) Recommendation

$$\boxed{\text{Have }\Phi\text{ verify it.}}$$

Not a second model. Not a subsequent paper.

This is a cheap, binary numerical audit using the existing Paper 9 codebase. It should be resolved before drafting §3 because it determines whether the κ→0 limit is a classical gradient-flow recovery or a projected/Clarke recovery.

My recommendation:

\boxed{\text{Run Φ first; only seek a second opinion if Φ finds interior nonzero derivatives.}}

That specific outcome would be serious:

\theta^\star\in(0,\pi/2),\quadc_A\neq c_B,\quad|c'(\theta^\star)|\gg 10^{-6}.

Then either

\mathcal F=-\mathcal C_{\min}

is not the right functional, or Paper 9’s recorded \theta^\star is not actually the argmax of c.

Bottom line

My ranking is:

\boxed{\text{Q2} > \text{Q1} > \text{Q4} > \text{Q3} > \text{Q5}.}

With confidence levels:

```
\[
 \boxed{
 \begin{array}{c|c|l}
 \text{Rank} & \text{Question} & \text{Confidence}\\
 \hline
 1 & Q2 & 92\%\\
 2 & Q1 & 84\%\\
 3 & Q4 & 74\%\\
 4 & Q3 & 61\%\\
 5 & Q5 & 43\%
 \end{array}
 }
 \]
```

The answer most ready for hostile review is Q2.

The answer most likely to be wrong is Q5, specifically its predicted outcome, not the audit design.